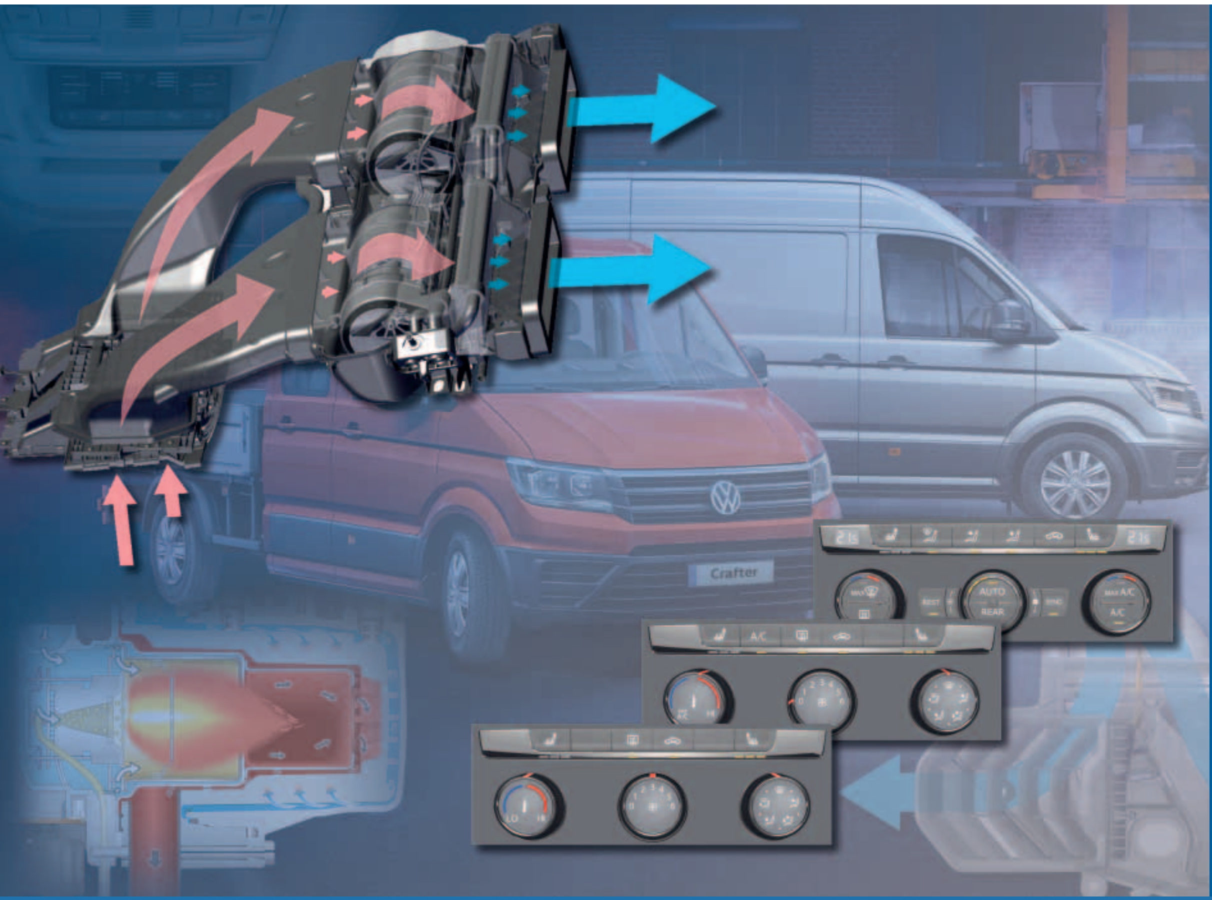
Service Training

Self-Study Programme 593

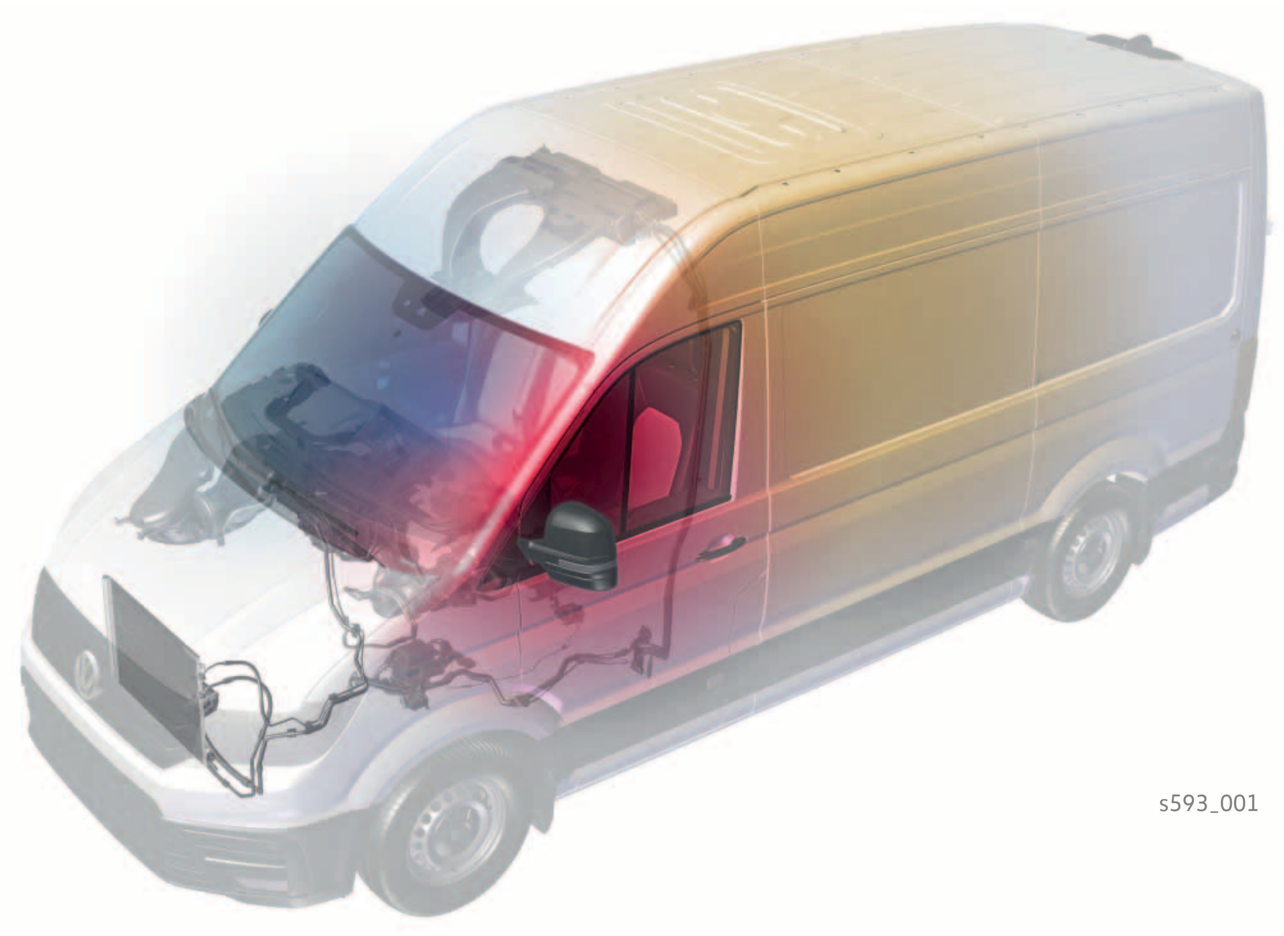
The Crafter 2017 heater and air conditioning system

Design and function



Hardly any vehicles nowadays do not have climate control, because it ensures pleasant temperatures in the vehicle. Climate control in vehicles not only serves the well-being of the vehicle occupants but also safety aspects, such as fog-free windows or the driver's ability to concentrate.
The climate control, which has been adapted to the extensive model range of the Crafter 2017, satisfies the various comfort requirements and safety aspects.

In this Self-Study Programme, the variants of climate control and the structure and function of the components involved are explained in more detail. Furthermore, information is provided about the various supplementary equipment items in the Crafter.



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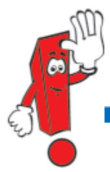
**The Self-study Programme shows the design and function of new developments.
The contents will not be updated.**

For current testing, adjustment and repair instructions, refer to the relevant service literature.



Important Note

Contents

Introduction

Climate control versions

The following different versions of the climate control system are available for the Crafter 2017:

- Heating and ventilation system
- Manual air conditioning system
- Automatic air conditioning system "Climatronic"

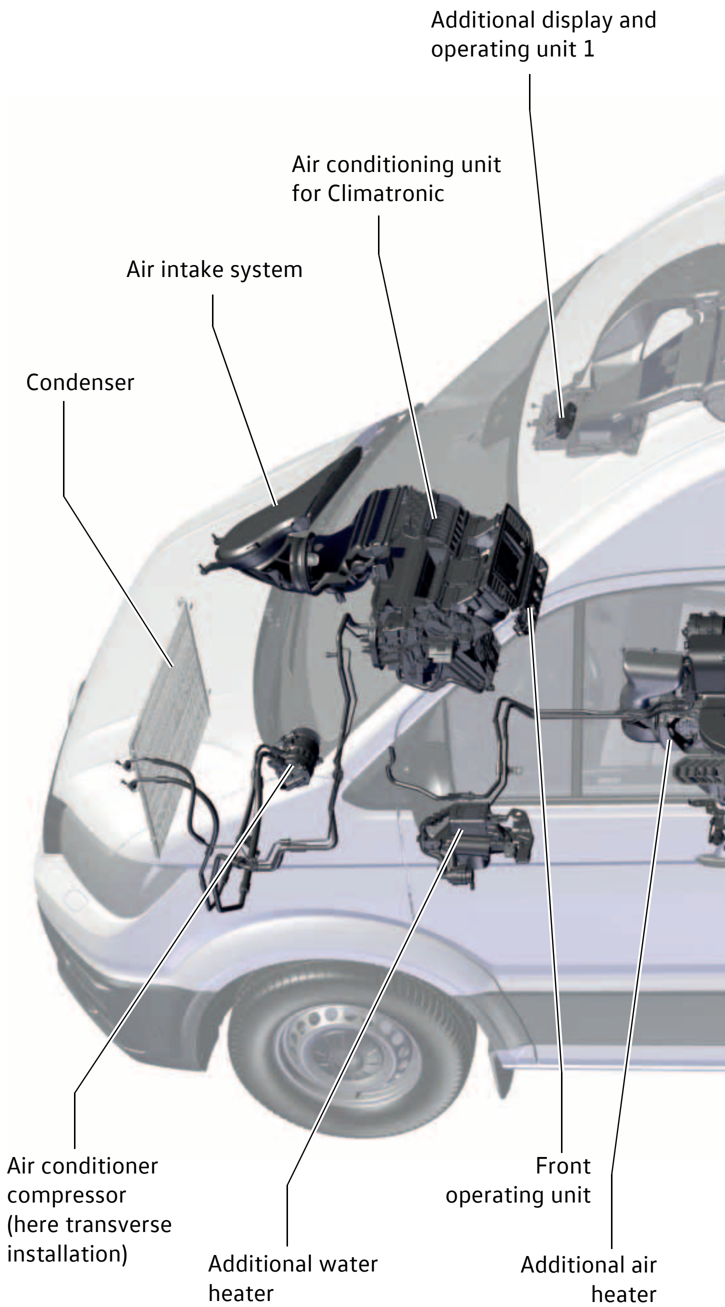
The heating and ventilation system is the basic equipment in the Crafter.

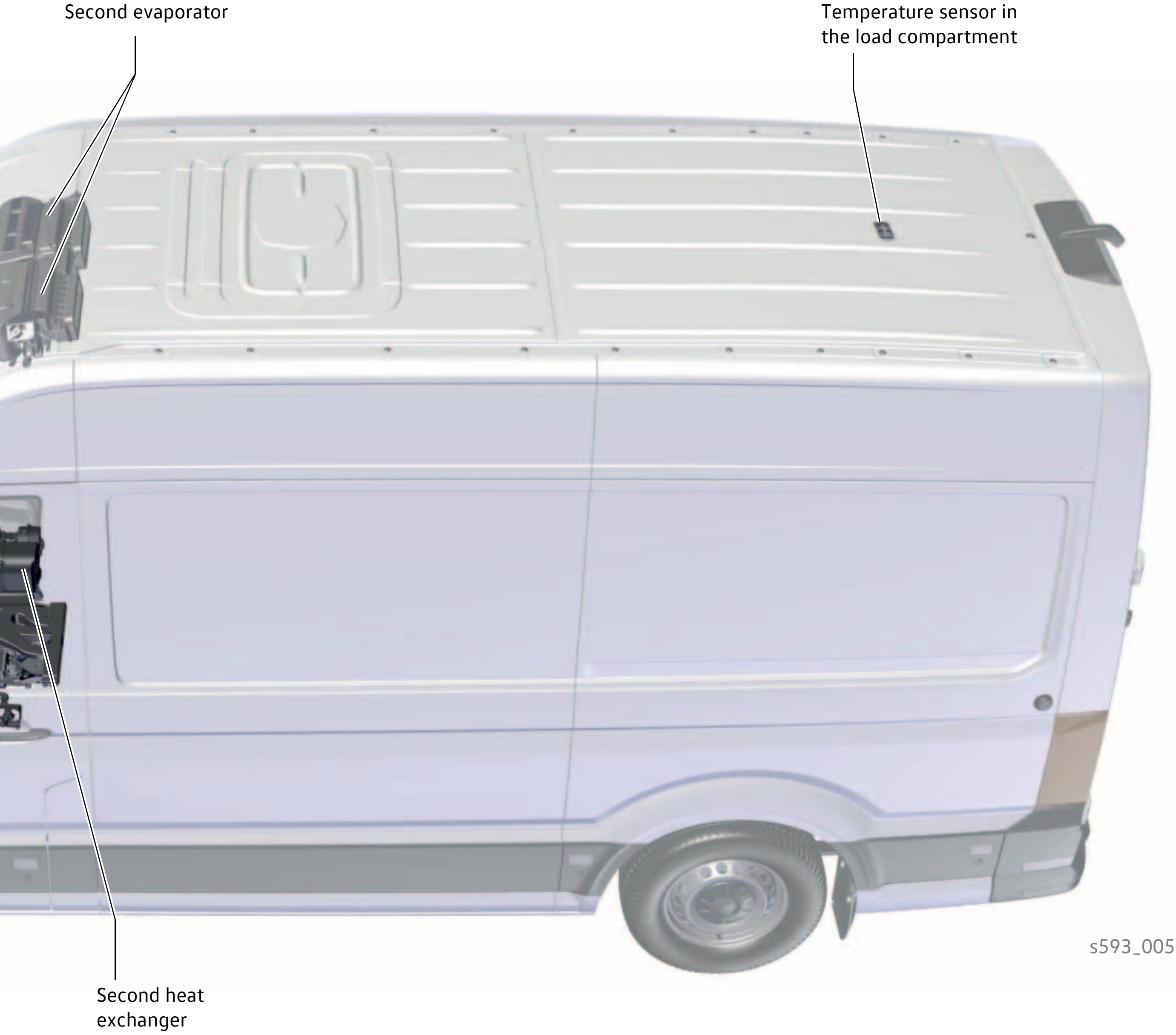
Supplementary equipment

Additional components can be fitted depending on equipment:

- Second evaporator for loadspace/passenger compartment
- Second heat exchanger for loadspace/passenger compartment
- Additional water heater
- Additional air heater (fuel-operated)
- Additional air heater (PTC in air conditioner)

The figure shows the maximum possible range of equipment.




Second evaporator
Temperature sensor in the load compartment
Second heat exchanger
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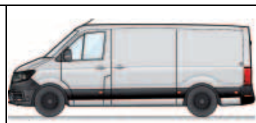
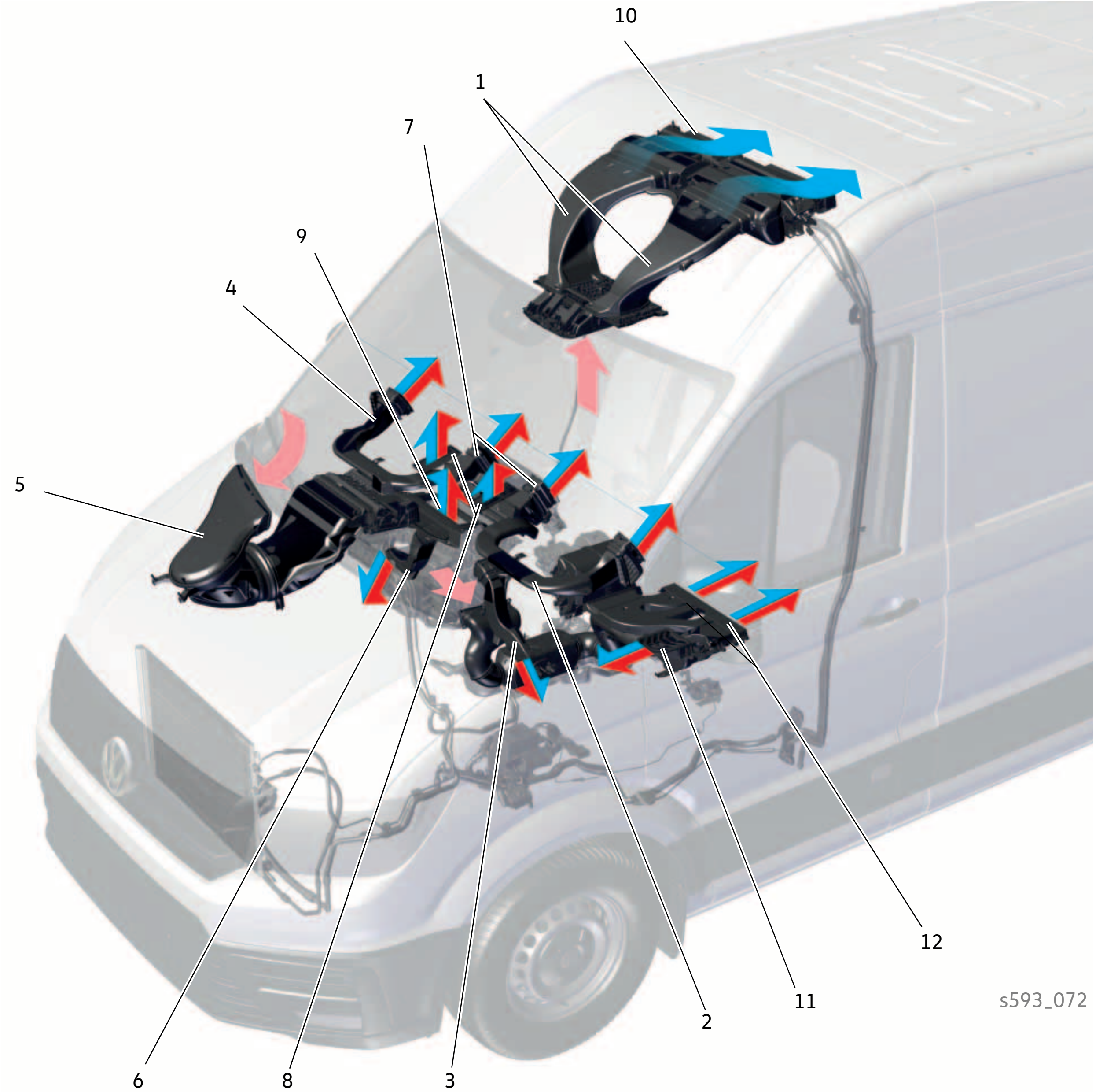
The climate control combinations

Type	 Drop-side vehicle with single cab	 Drop-side vehicle with double cab	 Panel van
Climate control versions			
Heating and ventilation system	●	●	●
Manual air conditioning system	❍	❍	❍
Automatic air conditioning system "Climatronic"	❍	❍	❍
Supplementary equipment			
Second evaporator 	–	–	❍ if air conditioning system is installed
Second heat exchanger 	–	–	❍
Additional water heater as auxiliary heater 	❍	❍	❍
Additional water heater as supplementary heater	❍	❍	❍
Additional air heater 	❍	❍	❍
Additional air heater (PTC) 	❍	❍	❍

● Standard ❍ Optional – Not available

Air distribution



Key

1 Air duct at the front on the second evaporator housing
2 Air duct for left dash panel vent
3 Driver side footwell vent
4 Air duct for right dash panel vent
5 Air intake duct
6 Front passenger side footwell vent
7 Centre vent
8 Top centre vent
9 Defrost function vent
10 Vents in the air conditioning box
11 Vents in the front area (additional air heater)
12 Vent for the loadspace/passenger compartment (heat exchanger, additional water heater or additional air heater)

The climate zones

The climate zones in the Crafter 2017 can be divided up as follows:

One zone

s593_007

Two zones left/right

s593_009

The entire vehicle interior is combined to form one climate zone. This can be done via the:

- Heating and ventilation system
- Manual air conditioning system

The vehicle interior is divided into two climate zones, namely the driver's cab at the front right and front left. This means that temperature requirements can be set independently for the driver and front passenger side. This can be done via the:

- Automatic air conditioning system “Climatronic”

Two zones front/rear

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The vehicle interior is divided into two climate zones, namely the driver's cab at the front and the loadspace/passenger compartment at the rear.
This can be done via the:

- Manual air conditioning system with a:
  - Second evaporator
  - Second heat exchanger
  - Second evaporator and second heat exchanger
- Heating and ventilation system with a:
  - Second heat exchanger

Three zones

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The vehicle interior is divided into three climate zones, namely the driver's cab at the front right and left, and the loadspace/passenger compartment at the rear. This can be done via the:

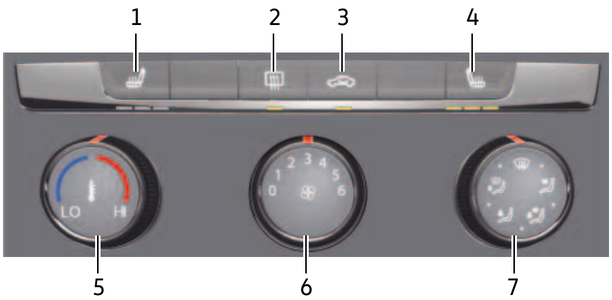
- Automatic air conditioning “Climatronic” with a:
  - Second evaporator
  - Second heat exchanger
  - Second evaporator and second heat exchanger

Climate control versions

The heating and ventilation system

Operating unit

The operating and control unit of the heating and ventilation system has three dials and one row of buttons. The system operates both manually and electrically. The temperature flap and the air distribution flaps are controlled purely electrically. The temperature is controlled manually on the operating unit.



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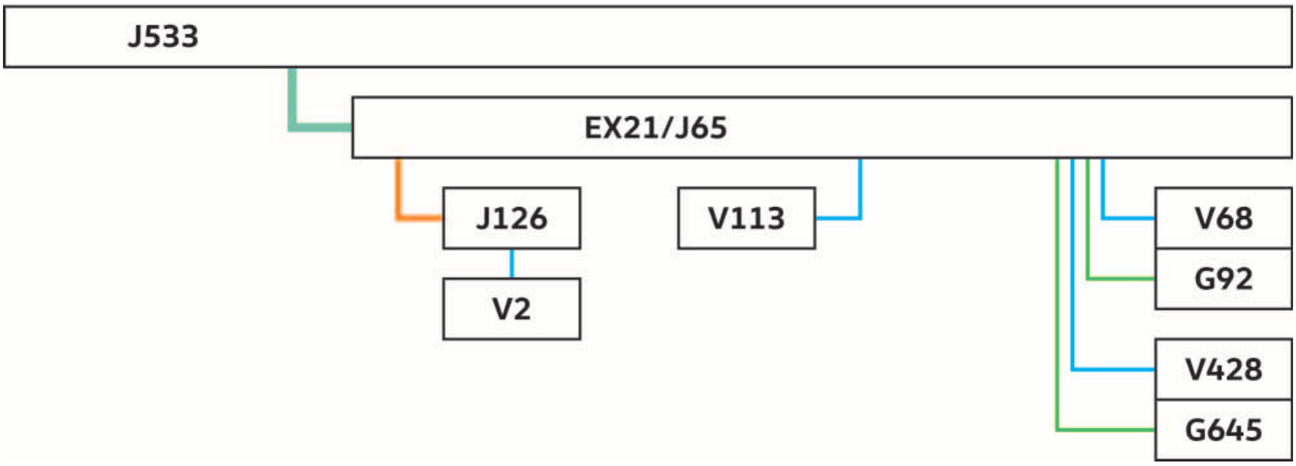
Key

1	Button for left seat heating
2	Button for heated rear window
3	Button for air recirculation mode
4	Button for right seat heating
5	Rotary knob for temperature selection
6	Rotary knob for fresh air blower
7	Rotary knob for air distribution

Networking



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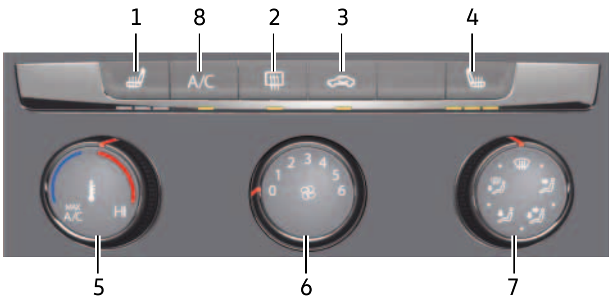
Key

EX21	Heater and air conditioning controls
G92	Potentiometer for temperature flap control motor
G645	Potentiometer for air distribution
J65	Heating control unit
J126	Fresh air blower control unit
J533	Data bus diagnostic interface
V2	Fresh air blower
V68	Control motor for temperature flap
V113	Control motor for air recirculation flap
V428	Front air distribution flap control motor
(teal line)	Convenience CAN bus
(orange line)	LIN bus wire
(green line)	Sensor line
(blue line)	Actuator line

The manual air conditioning system

Operating unit

The operating and control unit of the manual air conditioning system is identical in construction to the operating and control unit of the heating and ventilation system. However, it has been extended by the A/C button which switches the air conditioning system on and off. The control motors of the air and temperature flaps are electrically controlled by the operating unit. The temperature is controlled manually via the operating elements.



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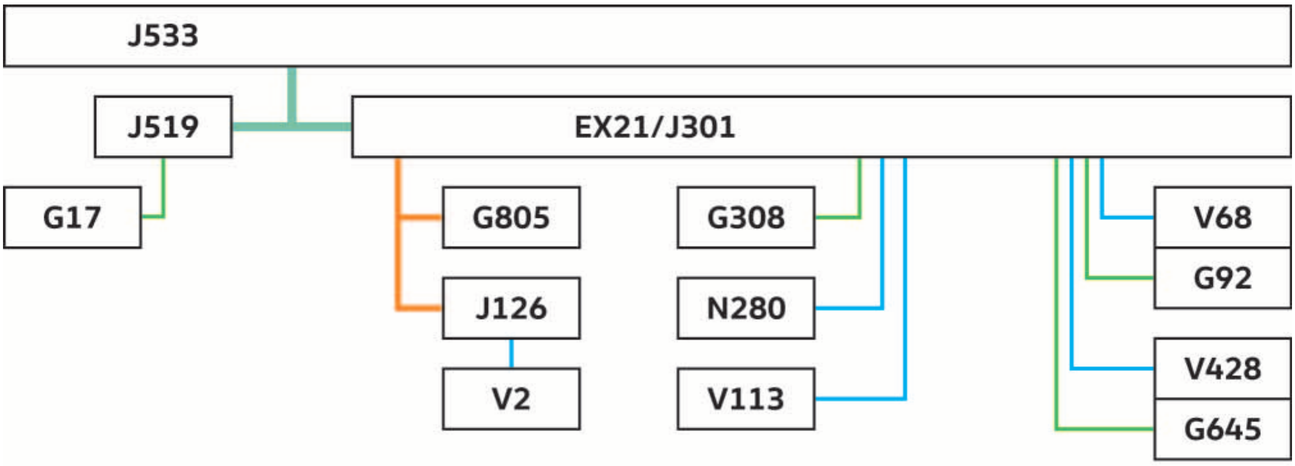
Key

1 Button for left seat heating
2 Button for heated rear window
3 Button for air recirculation mode
4 Button for right seat heating
5 Rotary knob for temperature selection
6 Rotary knob for fresh air blower
7 Rotary knob for air distribution
8 A/C button for switching the air conditioning system on and off

Networking



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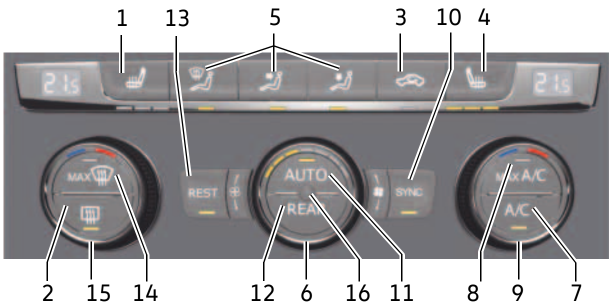
Key

EX21 Heater and air conditioning controls
G17 Exterior temperature sensor
G92 Potentiometer for temperature flap control motor
G308 Evaporator temperature sender
G645 Potentiometer for air distribution
G805 Pressure sender for refrigerant circuit
J126 Fresh air blower control unit
J301 Air conditioning control unit
J519 Onboard supply control unit
J533 Data bus diagnostic interface
N280 Air conditioner compressor regulating valve
V2 Fresh air blower
V68 Control motor for temperature flap
V113 Control motor for air recirculation flap
V428 Front air distribution flap control motor
Convenience CAN bus
LIN bus wire
Sensor line
Actuator line

"Climatronic", the automatic air conditioning system

Operating unit

The operating and control unit of the "Climatronic" automatic air conditioning system can be used to set temperature requirements independently of one another for the respective climate zones. The blower stages and air and temperature flaps are regulated automatically.



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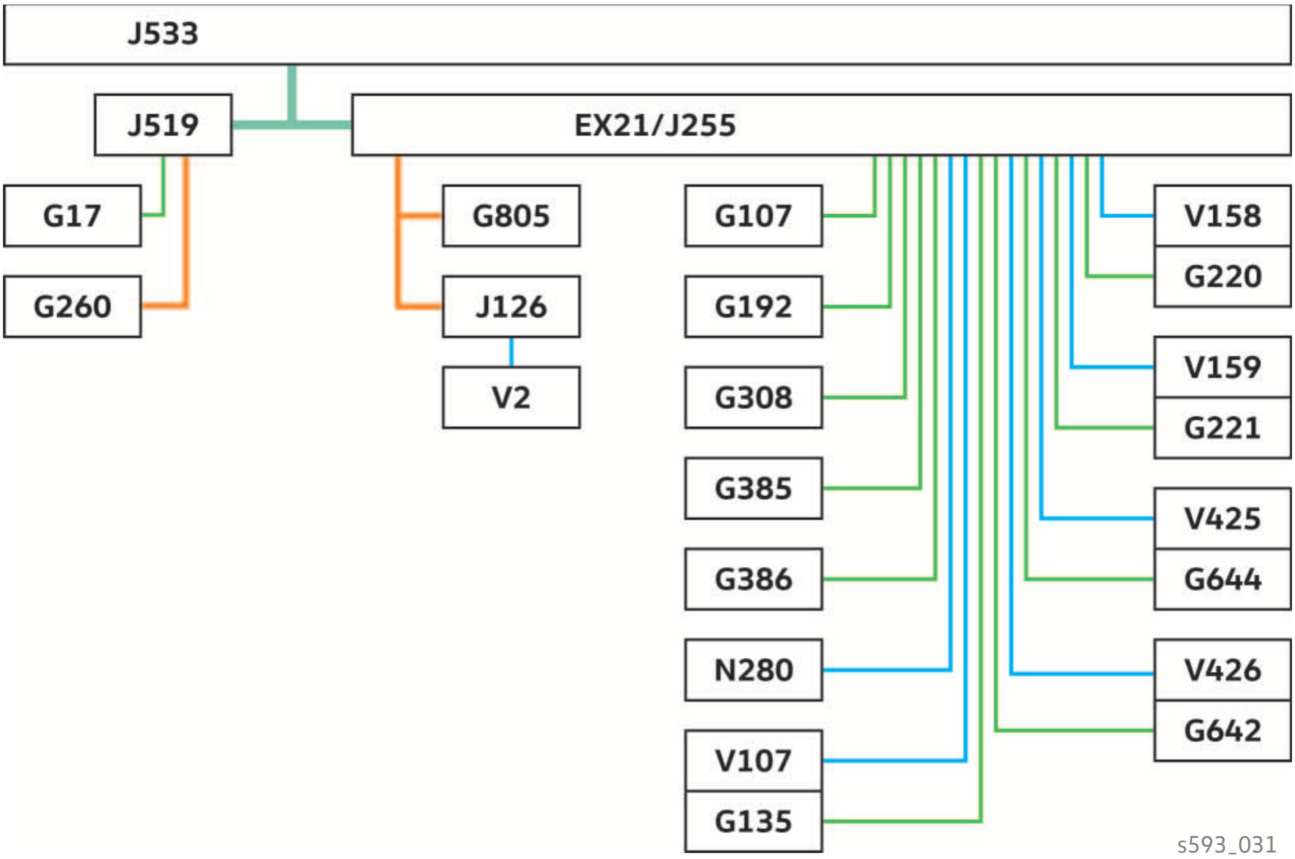
Key

1 Button for left seat heating
2 Button for heated rear window
3 Button for air recirculation mode
4 Button for right seat heating
5 Button for air distribution
6 Rotary knob for adjusting the blower level
7 A/C button for switching the air conditioning system on and off
8 Temperature setting to "LO", maximum blower output, air distribution to the chest vents
9 Rotary knob for adjusting the passenger side climate zone temperature
10 Synchronisation of the climate zones to the driver's value
11 Automatic control of the blower, temperature and air distribution depending on the intensity of the sunlight, outside and inside temperatures
12 Air conditioning regulation for passenger compartment
13 Use of residual heat when the engine is warm and ignition switched on, used for keeping the interior of the vehicle warm
14 Defrost, windscreen heating as well as maximum blower setting and air distribution towards the windscreen
15 Rotary knob for adjusting the driver side climate zone temperature
16 Interior temperature sensor

Networking



Key

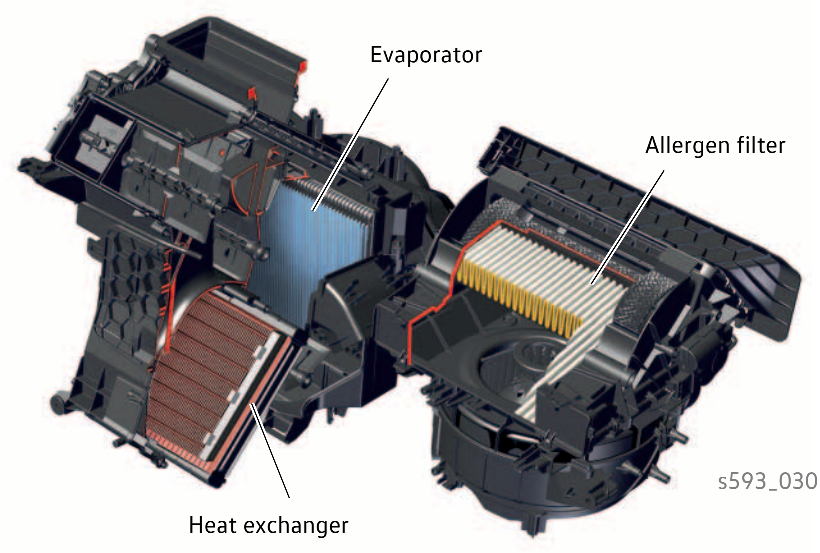
EX21 Heater and air conditioning controls
G17 Exterior temperature sensor
G107 Sunlight penetration photosensor
G135 Potentiometer for defroster flap control motor
G192 Footwell vent temperature sender
G220 Potentiometer for left temperature flap control motor
G221 Potentiometer for right temperature flap control motor
G260 Humidity sender for air conditioning system
G308 Evaporator temperature sender
G385 Front left chest vent temperature sensor
G386 Front right chest vent temperature sensor
G642 Potentiometer for front air distribution flap control motor
G644 Potentiometer for fresh air/air recirculation, air flow flap control motor
G805 Pressure sender for refrigerant circuit
J126 Fresh air blower control unit
J255 Climatronic control unit
J519 Onboard supply control unit
J533 Data bus diagnostic interface
N280 Air conditioner compressor regulating valve
V2 Fresh air blower
V107 Defroster flap control motor
V158 Control motor for left temperature flap
V159 Control motor for right temperature flap
V425 Fresh air/air recirculation, air flow flap control motor
V426 Control motor for air distribution
Convenience CAN bus
LIN bus wire
Sensor line
Actuator line

The heater and air conditioning unit

The heater and air conditioning unit is the main component in air distribution and temperature control. All climate control versions are based on a basic air conditioning unit. There are differences between the climate control versions in the number of flaps, control motors and temperature sensors.

The following diagram shows the maximum equipment of a heater and air conditioning unit for the 2-zone Climatronic with integrated evaporator, heat exchanger and allergen filter.

Design



Heat exchanger

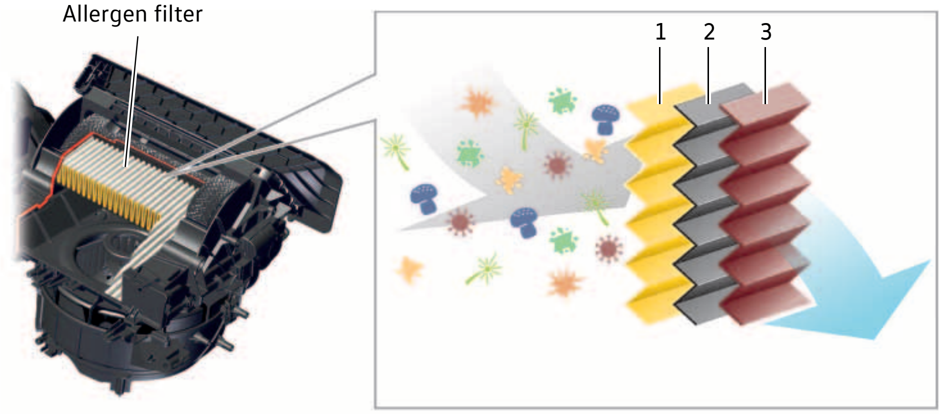
The heat exchanger is located in the air conditioning unit and is part of the cooling circuit.

Function

The heat exchanger generates warm air for the vehicle interior. The tubes and fins of the heat exchanger absorb heat from the cooling circuit of the engine. The air flowing through the heat exchanger is heated up and is directed into the vehicle interior.

Air Care (allergen filter)

The “Climatronic” automatic air conditioning system is equipped with the “Air Care” allergen filter.
A polyphenol coating has been added to the dust and pollen filter with activated charcoal.
This coating is a natural product that has anti-inflammatory properties and occurs in many plants.
It absorbs allergens and renders them harmless. The yellow-coloured coating makes the filter easy to distinguish from conventional dust and pollen filters.



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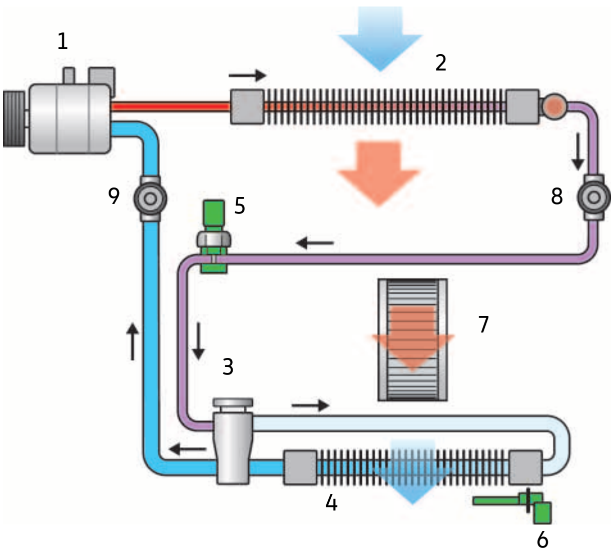
Key

1 Fleece layer with antibacterial and antiallergenic polyphenol coating
2 Activated charcoal layer to filter out odours and gases
3 Fleece layer to filter out pollen and dust

The structure of the refrigerant circuit

The refrigerant circuit of the air conditioning system in the Crafter basically consists of the following main components: a compressor, an evaporator, an expansion valve and a condenser. Pressure and temperature in the refrigerant circuit are monitored by sensors.

Schematic view



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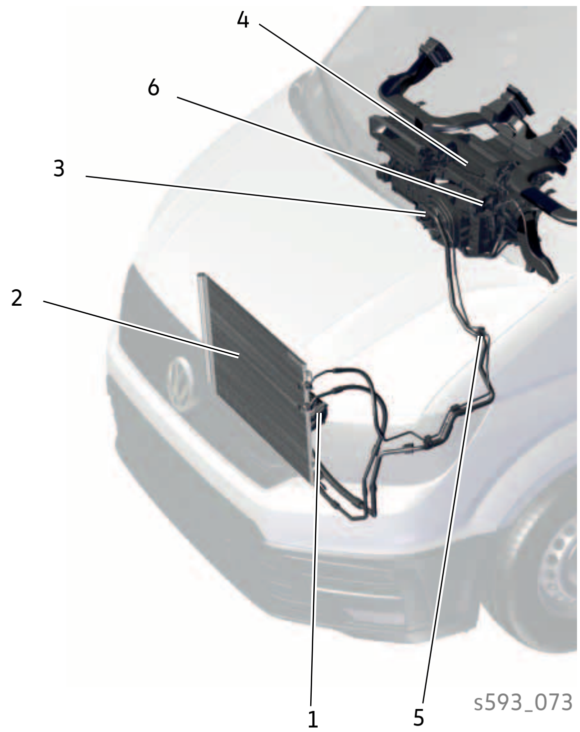
in the Crafter



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Key

1 Air conditioner compressor with control valve N280
2 Condenser with dryer cartridge
3 Expansion valve
4 Evaporator
5 Pressure sender for refrigerant circuit G805
6 Evaporator temperature sensor G308
7 Fresh air blower
8 High-pressure connection
9 Low-pressure connection

Gaseous refrigerant at high temperature and high pressure
Liquid refrigerant at medium temperature and high pressure
Liquid, partly gaseous refrigerant with a temperature well below ambient temperature and low pressure
Cold, gaseous refrigerant at low pressure

The components of the refrigerant circuit

Air conditioner compressor

s593_041

A 7-piston swash plate compressor operating on one side is used for compressing the refrigerant.
The compressor is mounted on the engine and is driven via a poly V-belt.
Due to the swash plate, the compressor has a variable displacement volume for adapting the compressor capacity to the refrigeration output requirement of the air conditioning system. The compressor is controlled via the external control valve for compressor N280.

Function

The compressor draws in cold, gaseous refrigerant at low pressure. It is compressed in the compressor.
The pressure increases and the temperature of the refrigerant also rises. In this phase, the refrigerant is gaseous and under high pressure at a high temperature. The compressor pushes the refrigerant to the condenser as a hot gas. The compressor therefore forms the separating point between the low-pressure and high-pressure sides of the refrigerant circuit.

Air conditioner compressor regulating valve N280

s593_099

The electric solenoid control valve is inserted into the compressor and is secured by a spring lock washer. It forms the interface between the low, high and crankcase pressure in the compressor and is a prerequisite for clutchless operation.
The swash plate is adjusted by these different pressures.

Function

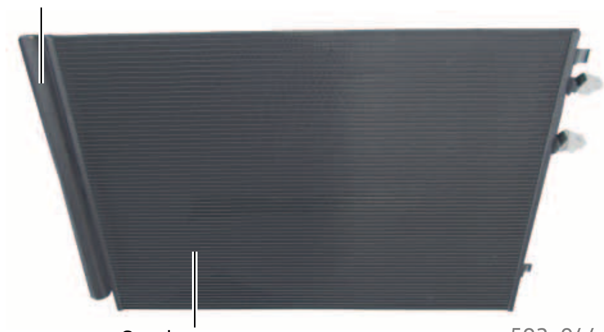
If, for example, a higher cooling capacity is required, the control valve is controlled by the control unit for Climatronic.
A pulse width-modulated voltage signal moves a plunger in the control valve. The duration of the applied voltage determines the adjustment travel. The adjustment changes the opening cross-section between the high pressure and the pressure in the crankcase of the compressor.
The crankcase pressure increases and causes a larger inclination of the swash plate via the piston stroke.

Refrigerant circuit

Condenser

The condenser is fitted in front of the radiator.

The fins of the condenser form a large heat exchange surface with good heat transfer.



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Function

The hot, gaseous refrigerant reaches the condenser. The fins of the condenser absorb heat. Cold outside air (airflow/fan) flows through the condenser. It absorbs heat. At the same time, the hot, gaseous refrigerant in the condenser cools down and becomes a liquid.
In this phase, the refrigerant is liquid and under high pressure at medium temperature.

Dryer cartridge

The dryer cartridge is located on the condenser.

Function

The dryer cartridge carries out various functions:

- **Absorption of moisture from the refrigerant circuit**
  Moisture that has penetrated the refrigerant circuit during installation, for example, is chemically bound.
- **Filtering the refrigerant**
  Suspended particles in the refrigerant, e.g. abrasion from the compressor, assembly dirt and the like, are filtered.
- **Expansion tank and reservoir**
  During different operating conditions, e.g. heat load on evaporator and condenser, compressor speed etc., different amounts of refrigerant are pumped through the circuit.
  The dryer cartridge serves to compensate for these fluctuations.

Expansion valve

s593_045

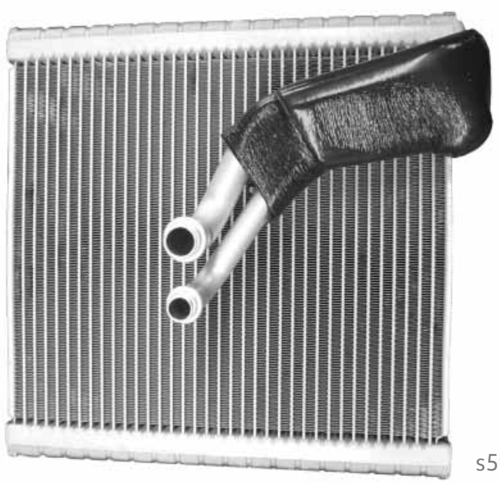
The expansion valve is located at the bulkhead of the vehicle and is arranged in the refrigerant circuit immediately before the evaporator.

The expansion valve regulates the refrigerant flow to the evaporator depending on the temperature of the refrigerant at the evaporator outlet.

Function

The pressure of the liquid refrigerant is reduced significantly in the expansion valve. The constriction inside the valve reduces the flow in the direction of the evaporator. The refrigerant can expand at the outlet of the expansion valve. The reduced pressure leads to partial evaporation of the refrigerant.
After expansion, the temperature of the refrigerant is significantly lower than the ambient temperature.
The expansion valve is the separating point between the high-pressure and low-pressure sections.

Evaporator



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The evaporator is part of the air conditioning unit. The fins of the evaporator form a large heat exchange surface with good heat transfer.

Function

In the evaporator, the refrigerant is at a temperature below the ambient temperature. It is vaporised.
The necessary evaporation heat is extracted from the air flow of the heater and air conditioning unit, which flows between the fins of the evaporator. This air flow, which is then directed into the passenger compartment, is cooled.
Cold, gaseous refrigerant at low pressure exits the evaporator and is sucked in by the compressor.

Sensors and actuators

System overview

The following system overview shows the sensors and actuators of the Climatronic.

Sensors

Ambient temperature sensor G17

Humidity sender for air conditioning system G260

Pressure sender for refrigerant circuit G805

Sunlight penetration photosensor G107

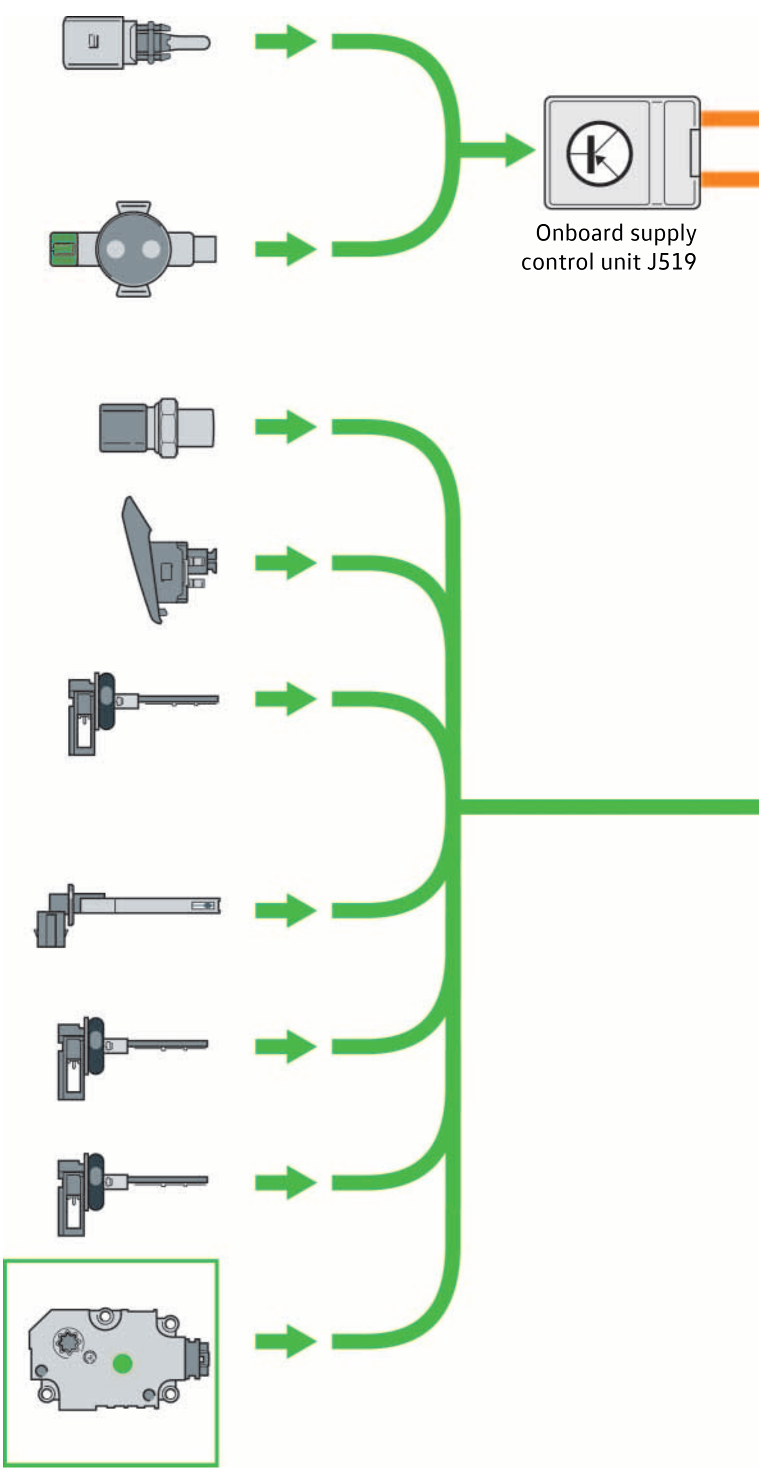
Footwell vent temperature sender G192

Evaporator temperature sensor G308

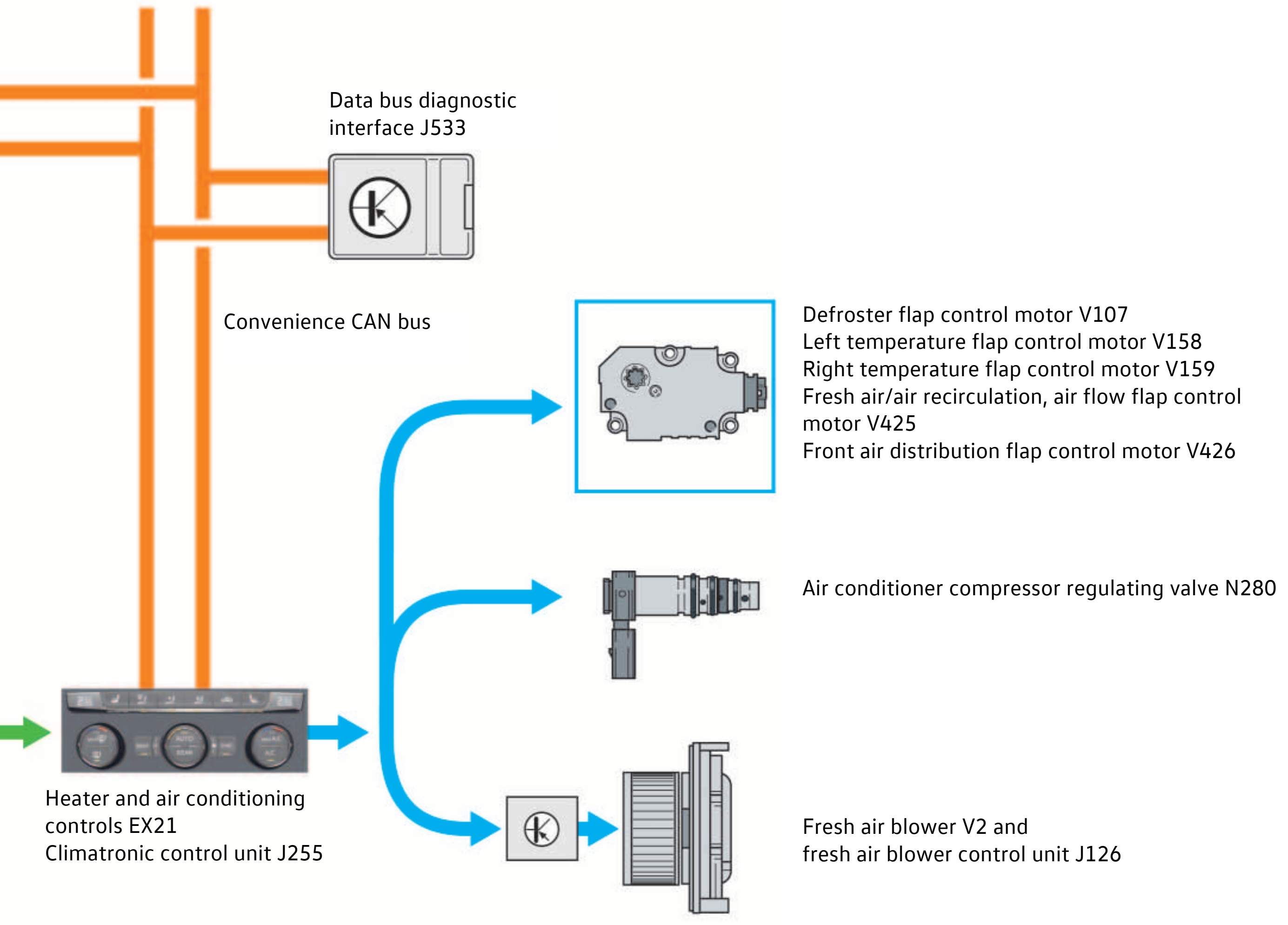
Front left chest vent temperature sensor G385

Front right chest vent temperature sensor G386

Potentiometer for defroster flap control motor G135
Potentiometer for left temperature flap control motor G220
Potentiometer for right temperature flap control motor G221
Potentiometer for front air distribution flap control motor G642
Potentiometer for fresh air/air recirculation, air flow flap control motor G644



Actuators



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The ambient temperature sensor G17

Fitting location

The ambient temperature sensor G17 is located behind the bumper at the front end.

Function

The temperature is measured by NTC thermistors. For regulating the air conditioning system, the temperature information is sent via the onboard supply control unit J519 to the control unit for air conditioning system J255.

s593_036

Effects of failure

The control unit for air conditioning system J255 assumes a fixed value of approx. 77 °C and the air conditioning system is switched off.

The humidity sender for air conditioning system G260

Fitting location

The humidity sender for air conditioning system G260 is located in the upper part of the windscreen together with the rain/light sensor.

Function

The humidity and the temperature at the windscreen are measured using a capacitive thin-layer sensor. The function of the sender is comparable to a parallel-plate capacitor. The measurement of the capacitance results in the degree of humidity.

The signals are sent to the onboard supply control unit and via the convenience CAN bus to the Climatronic control unit.

s593_037

Effects of failure

Without the sensor's signal, the control unit is no longer able to calculate the point in time from which moisture settles on the windows. The automatic defrost function fails.

The pressure sender for refrigerant circuit G805

s593_038

Fitting location

The pressure sender for refrigerant circuit G805 is installed in the high-pressure line between condenser and expansion valve.

Function

The pressure sender for refrigerant circuit G805 is connected directly to the climate control unit via LIN bus.
The signals are used to determine the actual refrigerant pressure in the refrigerant circuit and thus the required engine load.

Effects of failure

Without the pressure sender signal, the air conditioning system is switched off.

The sunlight penetration photosensor G107

s593_111

Fitting location

The photosensor is mounted on the dashboard in the middle behind the windscreen.

Function

The photosensor is affected by the solar radiation. It uses two photodiodes to determine the intensity of the sunlight and the position of the sun in relation to the vehicle. The information from the sensor is transmitted directly to the climate control unit.

Effects of failure

The control unit for air conditioning J255 adopts an assumed fixed value for solar radiation. The air conditioning system continues to work with this value.

The additional display and operating unit 1 E857

The additional display and operating unit 1 is installed in the roof console.

s593_033

If the Crafter is equipped with a heating and ventilation system or manual air conditioning system, the additional display and operating unit 1 E857 is always installed if the vehicle has one of the following supplementary equipment items:

- Second evaporator for loadspace/passenger compartment
- Second heat exchanger for loadspace/passenger compartment
- Additional water heater
- Additional air heater (fuel-operated)

On a Crafter with the “Climatronic” automatic air conditioning system, only the additional water heater and the additional air heater are operated via the additional display and operating unit 1.
The second evaporator and the second heat exchanger are operated via the REAR button on the operating and control unit (Heater and air conditioning controls EX21).

The settings for the additional climate controls are made using the following buttons on the additional display and operating unit 1.

Maximum equipment of the additional display and operating unit 1 E857

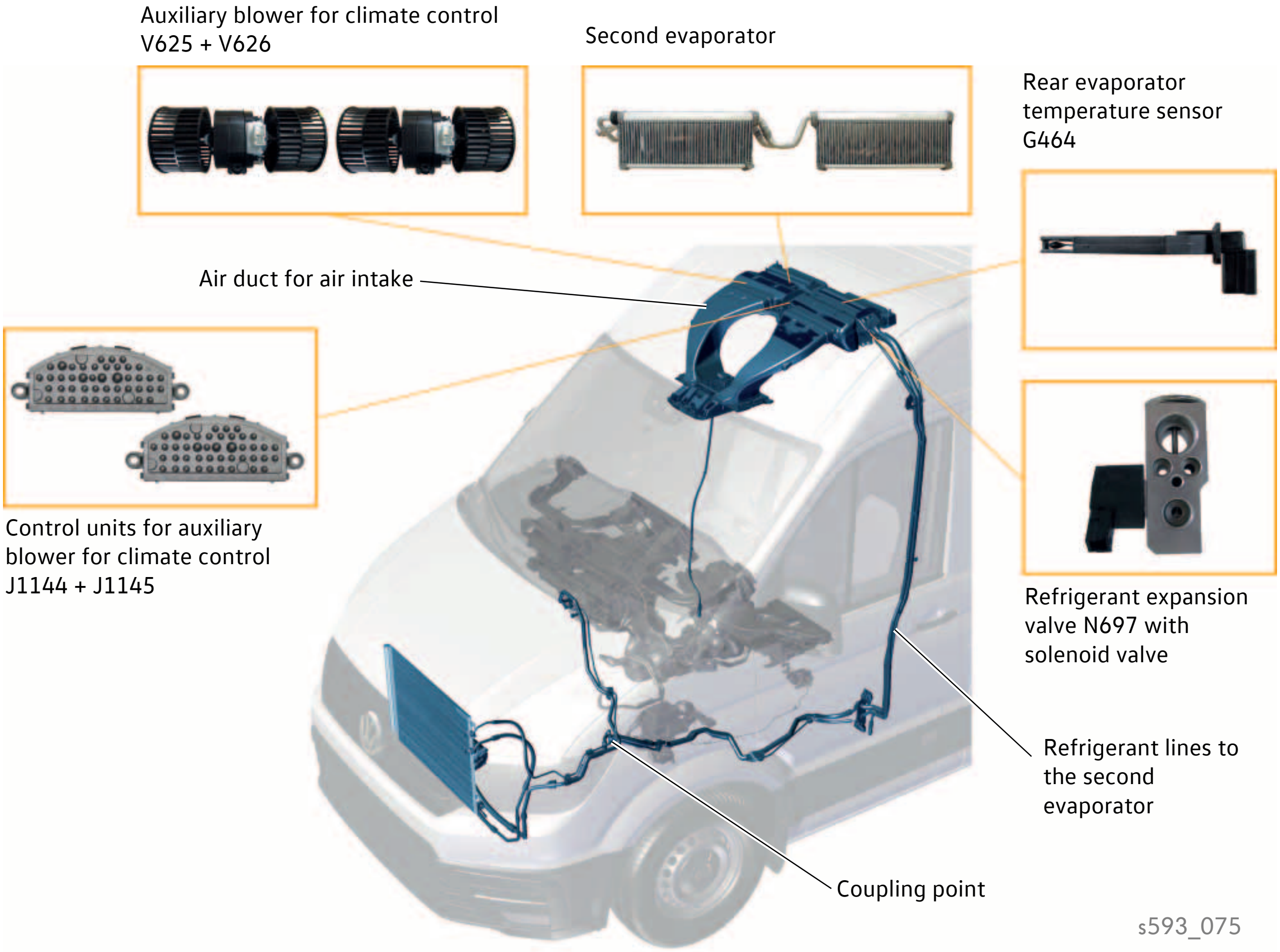
s593_035

The additional display and operating unit 1 is connected via a LIN data bus cable (LIN 2) to the Heater and air conditioning controls EX21.

The second evaporator

In vehicles with closed bodies, it may be sensible to extend the air conditioning system. For example, a second evaporator can optionally be installed above the driver's cab.

Fitting locations of components



The **second evaporator** is installed in the air conditioning box under the roof.
The **refrigerant lines** for the second evaporator are routed upwards in the B-pillar, thus providing the connection to the vehicle air conditioning system.
The **Rear evaporator temperature sensor G464** is installed in the left half of the air conditioning box on the evaporator.
The **expansion valve for refrigerant N697 with solenoid valve** blocks refrigerant entry at temperatures at the rear evaporator below 2 °C, with the evaporator blower at the rear switched off and the roof air conditioning system deactivated.
The blower speed control of the two **additional blowers for climate control V625 and V626** reduces or increases the outgoing cooled air flow.
This is controlled via the **control units for auxiliary blower for climate control J1144 and J1145**.

Refrigerant circuit

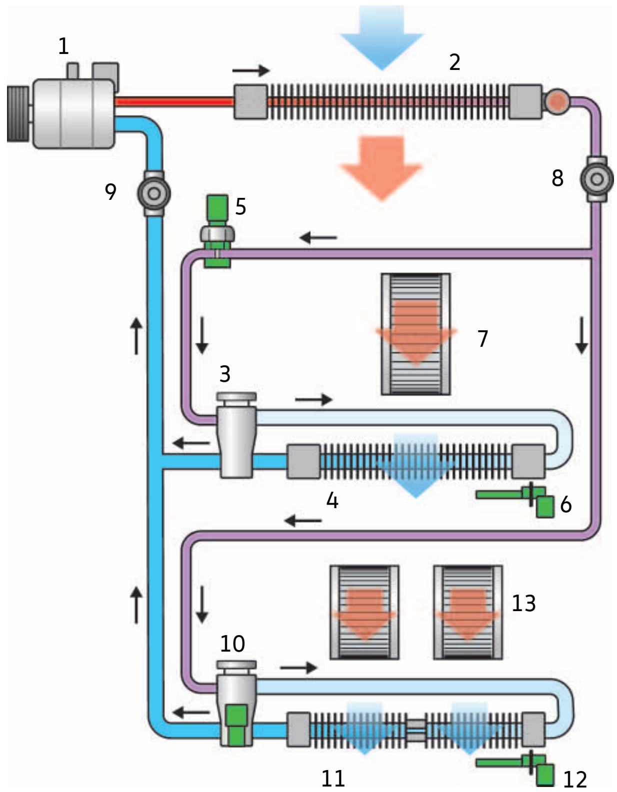
The refrigerant circuit of the vehicle is extended by the second evaporator, an expansion valve for refrigerant N697 with solenoid valve and a rear evaporator temperature sensor G464. The expansion valve relieves the pressure of the refrigerant and, in the evaporator, the refrigerant is completely vaporised, with the heat extracted from the air flowing through. The rear evaporator temperature sensor G464 monitors the temperature of the refrigerant at the evaporator. If this is below 2 °C, the solenoid valve on the expansion valve for refrigerant N697 blocks the refrigerant entry into the evaporator.

Key

1 Air conditioner compressor with control valve N280
2 Condenser with dryer cartridge
3 Expansion valve
4 Evaporator
5 Pressure sender for refrigerant circuit G805
6 Evaporator temperature sensor G308
7 Fresh air blower
8 High-pressure connection
9 Low-pressure connection
10 Refrigerant expansion valve, air conditioning system N697 with solenoid valve
11 Second evaporator
12 Rear evaporator temperature sensor G464
13 Auxiliary blower for air conditioning V625 and V626

- Gaseous refrigerant at high temperature and high pressure
- Liquid refrigerant at medium temperature and high pressure
- Liquid, partly gaseous refrigerant with a temperature well below ambient temperature and low pressure
- Cold, gaseous refrigerant at low pressure

Schematic view



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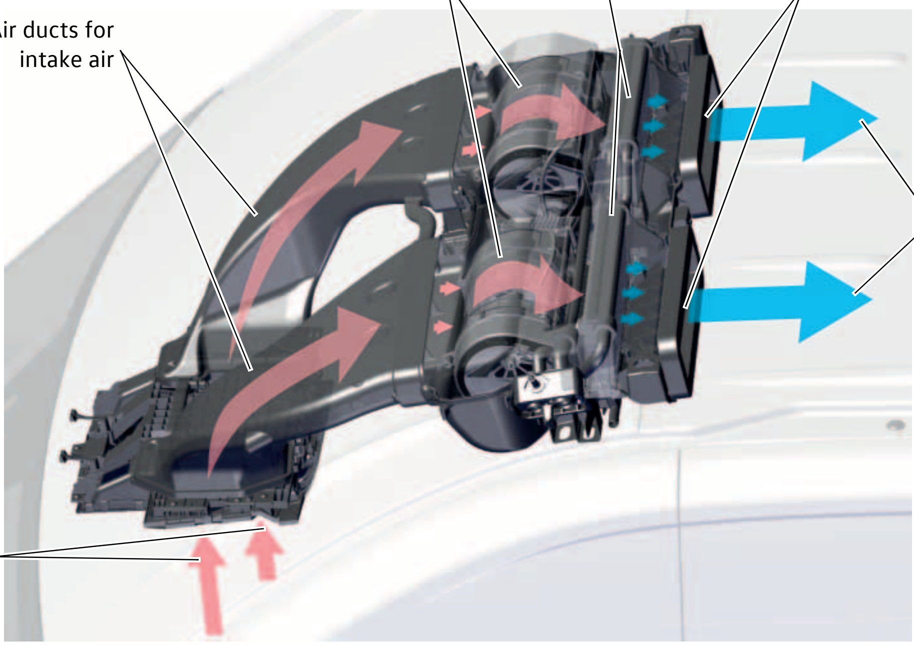
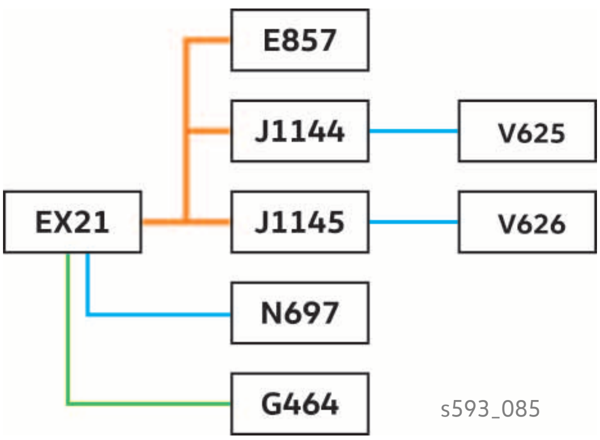
Air duct system

The air is sucked into the area of the windscreen by the auxiliary blower for climate control V625 and V626 and is channelled to the evaporator via two air ducts. The air cooled by the evaporator is blown directly to the rear of the vehicle via the vents in the air conditioning box. The speed control of the two auxiliary blowers for climate control reduces or increases the outgoing cooled air flow.



s593_082

Networking



s593_085

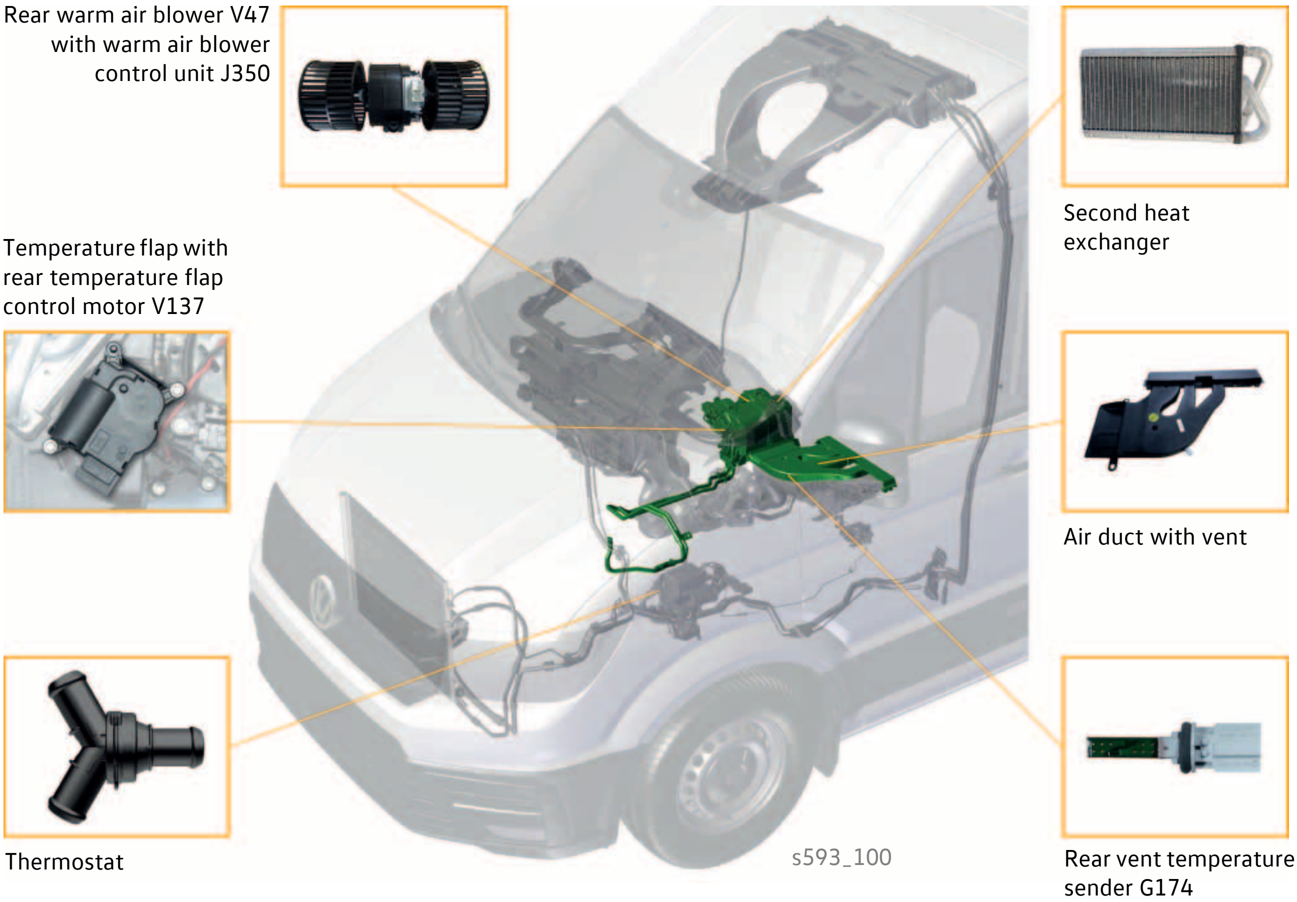
Key

E857	Additional display and operating unit 1
EX21	Heater and air conditioning controls
G464	Rear evaporator temperature sensor
J1144	Control unit for additional blower for climate control
J1145	Control unit for auxiliary blower 2 for climate control
N697	Refrigerant expansion valve, air conditioning system N697 with solenoid valve
V625	Auxiliary blower for climate control
V626	Auxiliary blower 2 for climate control
(orange line)	LIN bus wire (19.2 kbit/s)
(green line)	Sensor line
(blue line)	Actuator line

The second heat exchanger

A Crafter with closed body can optionally be extended with a second heat exchanger.
It enables the additional output of heat in the rear area of the vehicle.

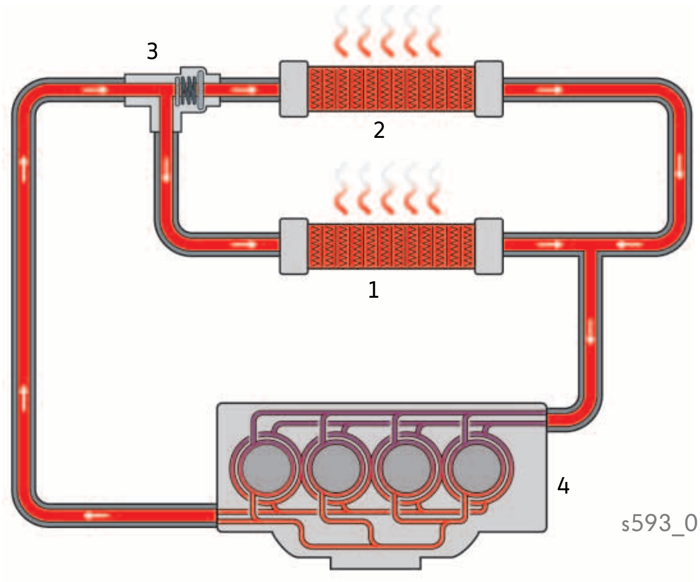
Fitting locations of components



The **second heat exchanger** is installed under the front passenger seat and integrated into the vehicle's coolant circuit. The **thermostat** is always installed with a second heat exchanger and is located on the underbody near the heater unit of the additional water heater. The speed of the **rear warm air blower V47** is controlled by the **warm air blower control unit J350**. The **rear temperature flap control motor V137** is infinitely variably controlled by the climate control unit.
The **rear vent temperature sender G174** is located in the air duct of the second heat exchanger. It is used to control the **temperature flap** in the second heat exchanger. From the air duct, the air is introduced into the loadspace/passenger compartment via the vents.

Coolant circuit

A thermostat is always installed in conjunction with a second heat exchanger. It opens the coolant circuit to the second heat exchanger from a coolant temperature of 55 °C and higher, and then lets approx. 25 percent of the preheated coolant flow through the second heat exchanger. The other approx. 75 percent of the preheated coolant flows on to the first heat exchanger.



s593_050

Key

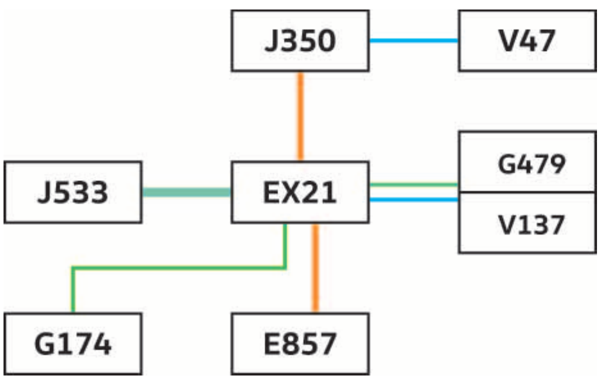
1 First heat exchanger

2 Second heat exchanger

3 Thermostat

4 Engine

Networking



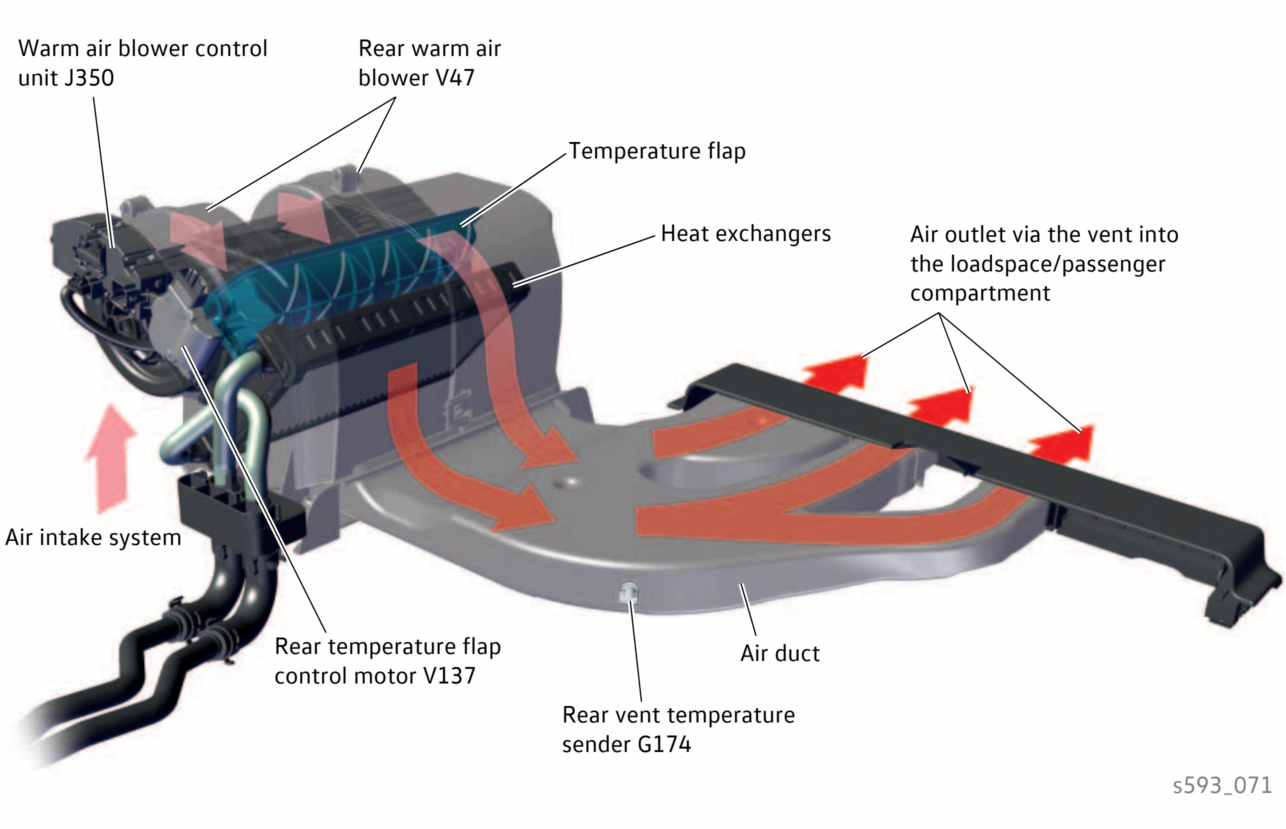
s593_108

Key

E857	Additional display and operating unit 1
EX21	Heater and air conditioning controls
G174	Rear vent temperature sender
G479	Potentiometer for rear temperature flap motor
J350	Warm air blower control unit
J533	Data bus diagnostic interface
V47	Rear warm air blower
V137	Rear temperature flap control
(teal line)	Convenience CAN bus (500 kbit/s)
(orange line)	LIN bus wire
(green line)	Sensor line
(blue line)	Actuator line

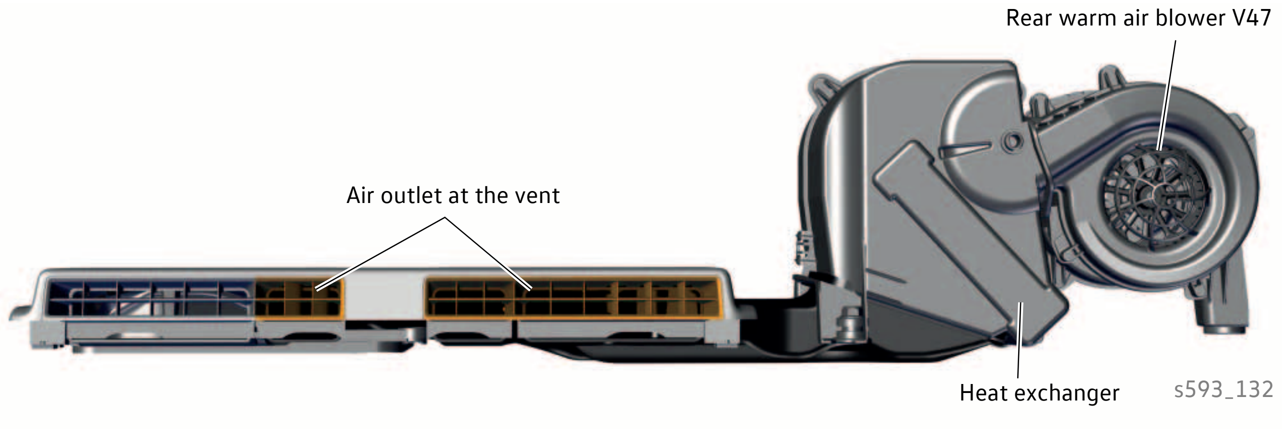
Air duct system

Cool air is sucked in from the interior of the vehicle by the rear warm air blower V47.
The speed of the rear warm air blower V47 is controlled by the warm air blower control unit J350.
The air is channelled via a temperature flap to the heat exchanger and from there via the air duct to the vent.



s593_071

Air is discharged from the vent into the loadspace/passenger compartment via the vent area marked in orange.

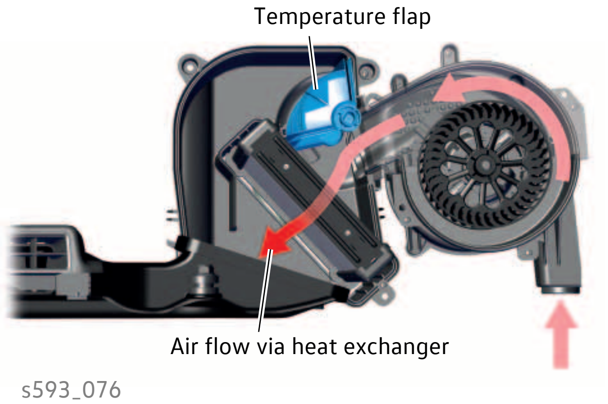


s593_132

Temperature flap

The temperature flap regulates how much air is drawn in and then flows through the heat exchanger. It can be infinitely variably adjusted by the rear temperature flap control motor V137. The control motor is controlled by the climate control unit.

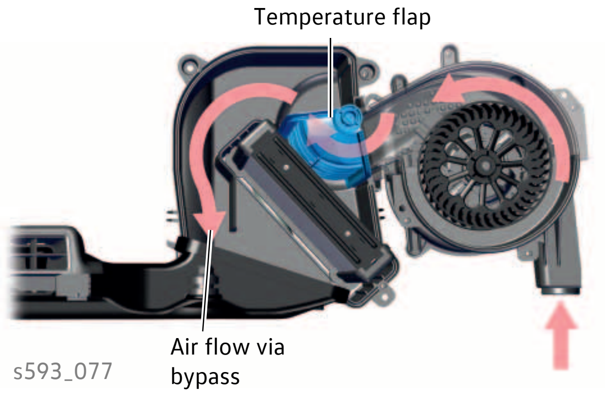
Temperature flap completely open



s593_076

If the temperature flap is completely open, the air drawn into the interior is channelled completely through the heat exchanger and warm air reaches the interior of the vehicle.

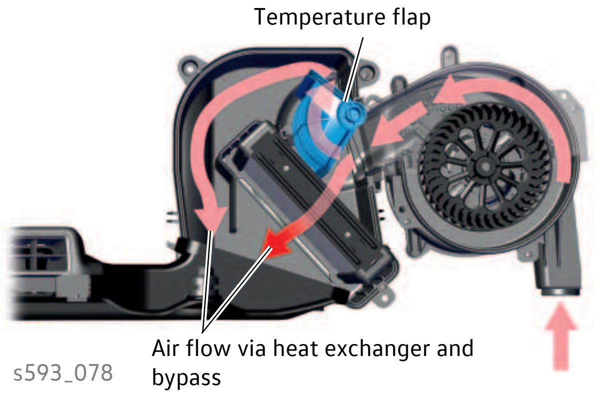
Temperature flap fully closed



s593_077

If the temperature flap is fully closed, the intake air is diverted past the heat exchanger via a bypass and the drawn-in cool air enters the interior of the vehicle.

Temperature flap between open and closed



s593_078

The temperature flap can be set between fully open and fully closed so that one part of warm air and one part of cool air always reaches the interior.

Additional water heater

The Crafter can be optionally equipped with an additional water heater from Webasto.
The additional water heater with the designation “Thermo Top Evo 5” can be used as a supplementary heater or as an auxiliary heater.

The **task of the auxiliary heater** is to pre-heat the vehicle interior without being reliant on the heat output by the running engine. This involves hot coolant flowing only through the heat exchanger while the internal combustion engine is not running. The heater coolant shut-off valve N279 blocks the supply to the combustion engine coolant circuit.

The **task of the supplementary heater** is to increase the heating output for vehicle engines with a high degree of efficiency and consequently low heat output. The additional water heater thus supports the heat output when the internal combustion engine is running.

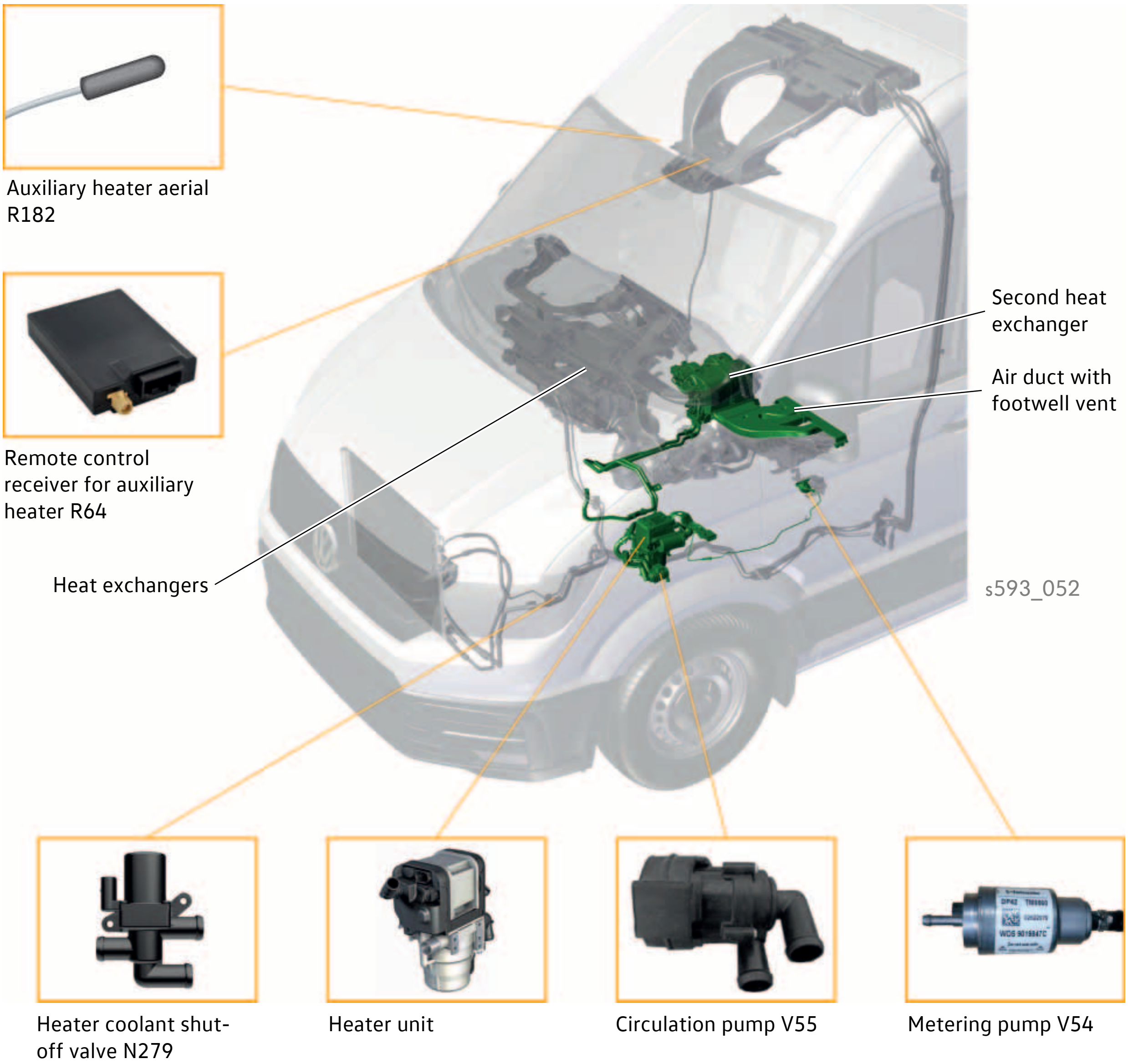
Technical data

Manufacturer	Webasto
Designation	Thermo Top Evo 5
Use	For use on Crafter panel vans, platform vans with single cab, platform vans with double cab, bodybuilder chassis
Fuel	Diesel as per EN 590
Heating output	2 regulation stages • Full load 5000 watts • Part load 2500 watts
Fuel consumption	approx. 0.6 l/h max.
Operating voltage	12 volt
Electrical power consumption	32 watts max.
Undervoltage disconnection	10.5 V > 20 seconds

s593_046

If the additional water heater is installed as an auxiliary heater, it can also be operated as a supplementary heater.

Fitting locations of components



The **heater** of the additional water heater with auxiliary heater control unit J364 is installed on the underbody at the left.
The **metering pump V54** is located in the middle of the underbody and supplies the additional water heater with the required fuel. It is a reciprocating piston pump and is controlled in a cyclical manner.
The **circulation pump V55** is located on the underbody close to the heater.
The **heater coolant shut-off valve N279** is screwed onto the bulkhead. It is a 3/2-way valve. In auxiliary heating mode, the valve blocks access to the coolant circuit of the internal combustion engine. This warms up the water faster and the interior is heated faster.
The **remote control receiver for auxiliary heating R64** is installed at the front of the roof near the interior mirror.

Heater unit system connections

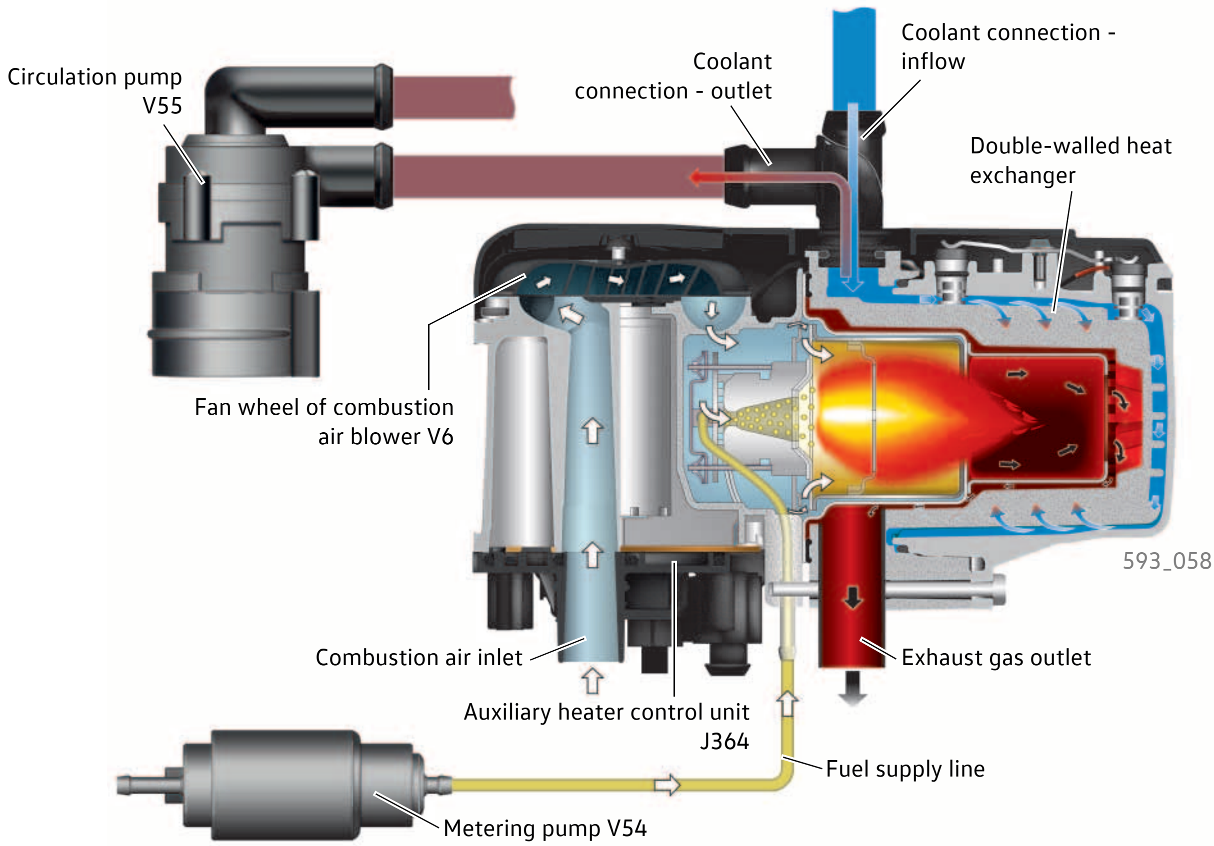
The heater is the central main component of the additional water heater. It has the following system connections:

Fuel system
Fuel is pumped by the metering pump V54 from the fuel tank to the heater via a separate line (fuel supply).

Coolant circuit
The coolant enters the double-walled heat exchanger via the coolant connection inlet. There, the coolant absorbs the heat generated by the combustion of the fuel. The heated coolant enters the coolant circuit via the coolant connection outlet. With the aid of the circulation pump V55, the heated coolant reaches the two heat exchangers via the coolant circuit.

Air supply
The air required for combustion is drawn in through the combustion air intake by the combustion air blower V6 in the heater unit.

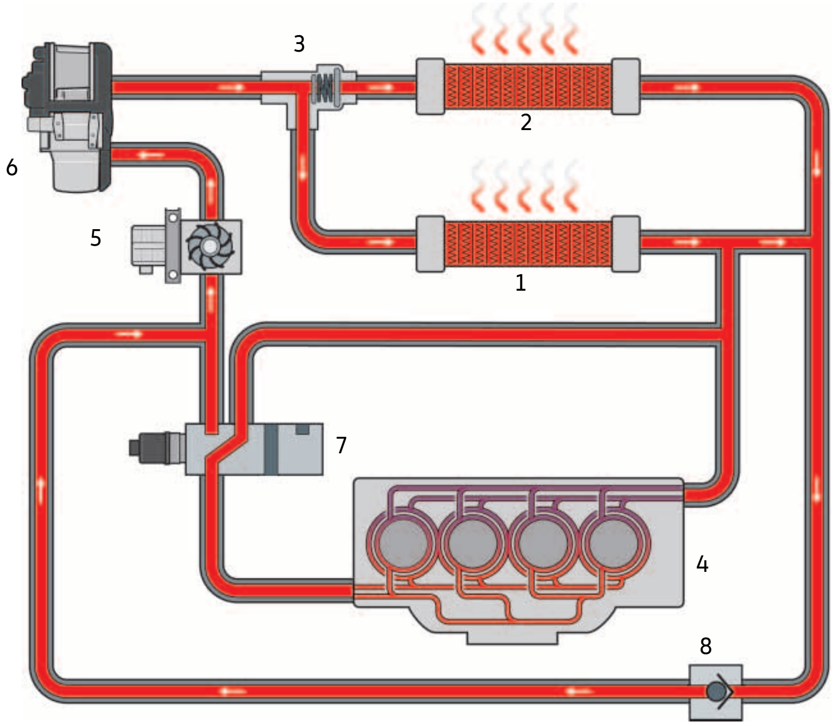
Vehicle electrical system
The additional water heater is connected to the electrical system via the electrical connections for the auxiliary heater control unit J364. The control unit is a component of the infotainment CAN bus.



593_058

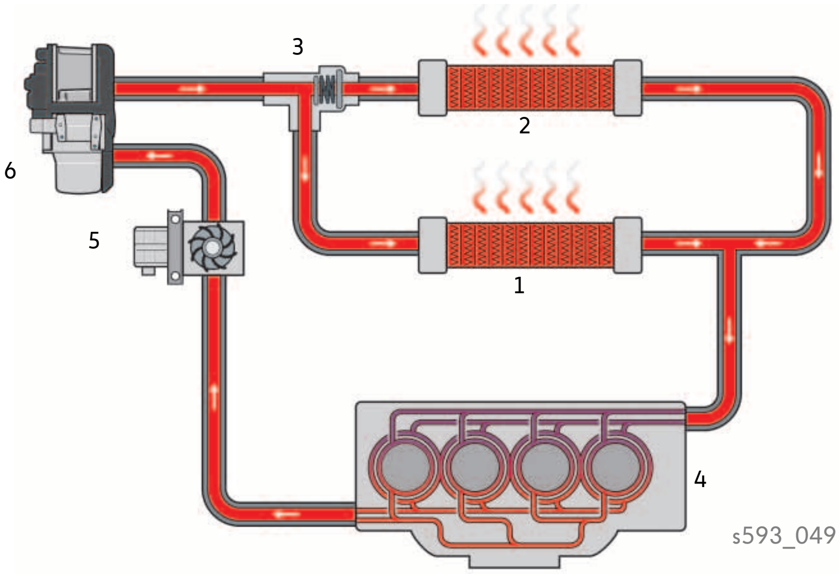
Coolant circuits

Auxiliary heating function with two heat exchangers



s593_048

Supplementary heating function with two heat exchangers

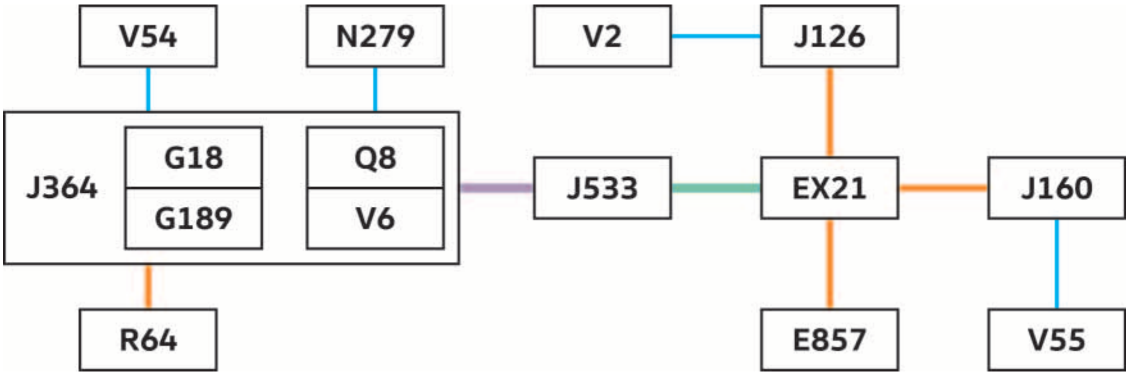


s593_049

Key

1 Heat exchangers
2 Second heat exchanger
3 Thermostat
4 Engine
5 Circulation pump V55
6 Additional water heater
7 Heater coolant shut-off valve N279
8 Non-return valve

Networking



s593_047

Key

E857	Additional display and operating unit 1
EX21	Heater and air conditioning controls
J126	Fresh air blower control unit
J160	Circulation pump relay
J364	Auxiliary heater control unit
J533	Data bus diagnostic interface
N279	Heater coolant shut-off valve
R64	Remote control receiver for auxiliary heater
V2	Fresh air blower
V54	Metering pump
V55	Circulation pump

Internal components of the heater

G18	Temperature sensor
G189	Overheating sensor
Q8	Glow plug with flame monitor
V6	Combustion air blower

Infotainment CAN bus (500 kbit/s)

Convenience CAN bus (500 kbit/s)

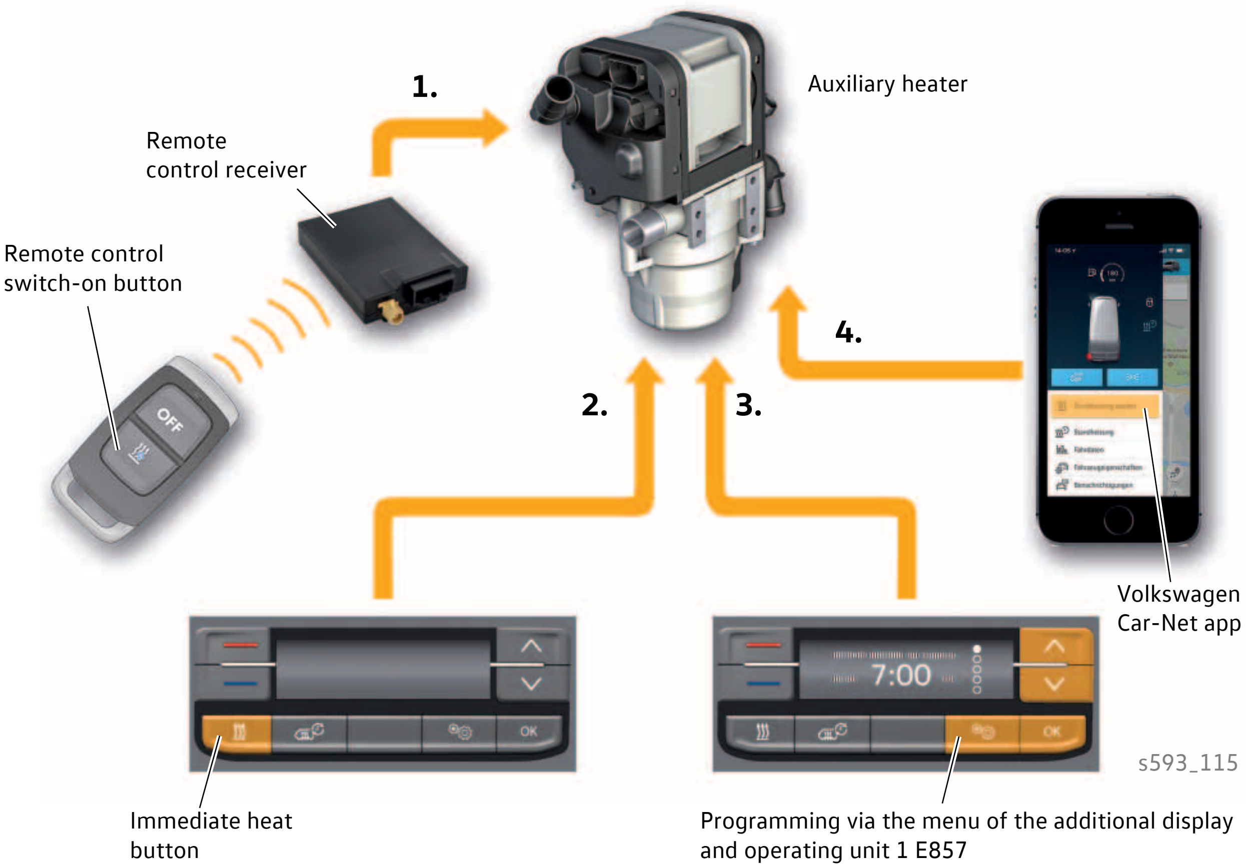
LIN bus wire

Actuator line

The climate control unit is the master for all functions in connection with the additional water heater.

Operation

The following variants are available for switching on the **auxiliary heater**:



1. Manually via the remote control
2. Manually by pressing the immediate heat button on the additional display and operating unit 1
3. Automatically via pre-programming in the menu of the additional display and operating unit 1
4. Manually via smartphone or web portal (only if equipped with Security & Service, Car-Net app and corresponding registration via customer portal)

In **supplementary heater** mode, the heater unit is not operated by the customer, but instead additionally heats the coolant when the internal combustion engine is running.
If the "Automatic supplementary heater" function is activated in the Infotainment system climate menu, the additional water heater is used as a supplementary heater when the following switch-on conditions are met:

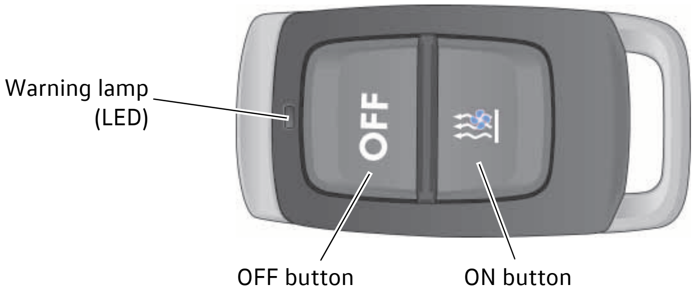
- Terminal 15 ON
- Ambient temperature measured value is not incorrect
- Ambient temperature less than approx. +5 °C before the supplementary heater is started
- Engine speed greater than 600 rpm
- No transport mode active
- No blocking mode active
(no low-fuel warning active, no undervoltage)
- No blocking status active
(no entry in event memory due to crash or overheating)

Supplementary equipment

Remote control

The “ON” button is pressed on the remote control. The remote control receiver for auxiliary heater R64 transmits the received signal via LIN bus to the auxiliary heater control unit J364. From there via the infotainment CAN bus to the gateway climate control unit and then via the convenience CAN to the climate control unit. The switch-on signal is now sent back to the auxiliary heater control unit and heating mode begins. After the start, the active operating mode appears on the display of the additional display and operating unit 1.

The auxiliary heater can be switched off again using the “OFF” button.



s593_083

Programming

The auxiliary heater can be programmed for three different departure times (timers) in the settings menu using the buttons on the additional display and operating unit 1. The switch-on time and operating duration of the auxiliary heater is then determined by the onboard supply control unit according to the ambient temperature and the required interior temperature.
When programming, it is important that the date and time in the vehicle are set correctly.

There is no serial function of the timer, i.e. after the timer has expired, it must be selected again in order to use the set function again.

You can program an operating duration between 10 minutes and 120 minutes in 10-minute steps. The set operating duration is only used for starting with the remote control or with the instant heating button.

The additional air heater

The Crafter can be optionally equipped with an additional air heater from Eberspächer.
The additional air heater with the designation “Airtronic M2 D4A” is operated as an auxiliary heater.
Under certain conditions it can be switched on by the climate control unit as a supplementary heater.

The **auxiliary heater** has the task of heating the vehicle interior. When the internal combustion engine is at a standstill, cool room air is drawn in by the additional air heater, heated and then delivered to the vehicle interior.

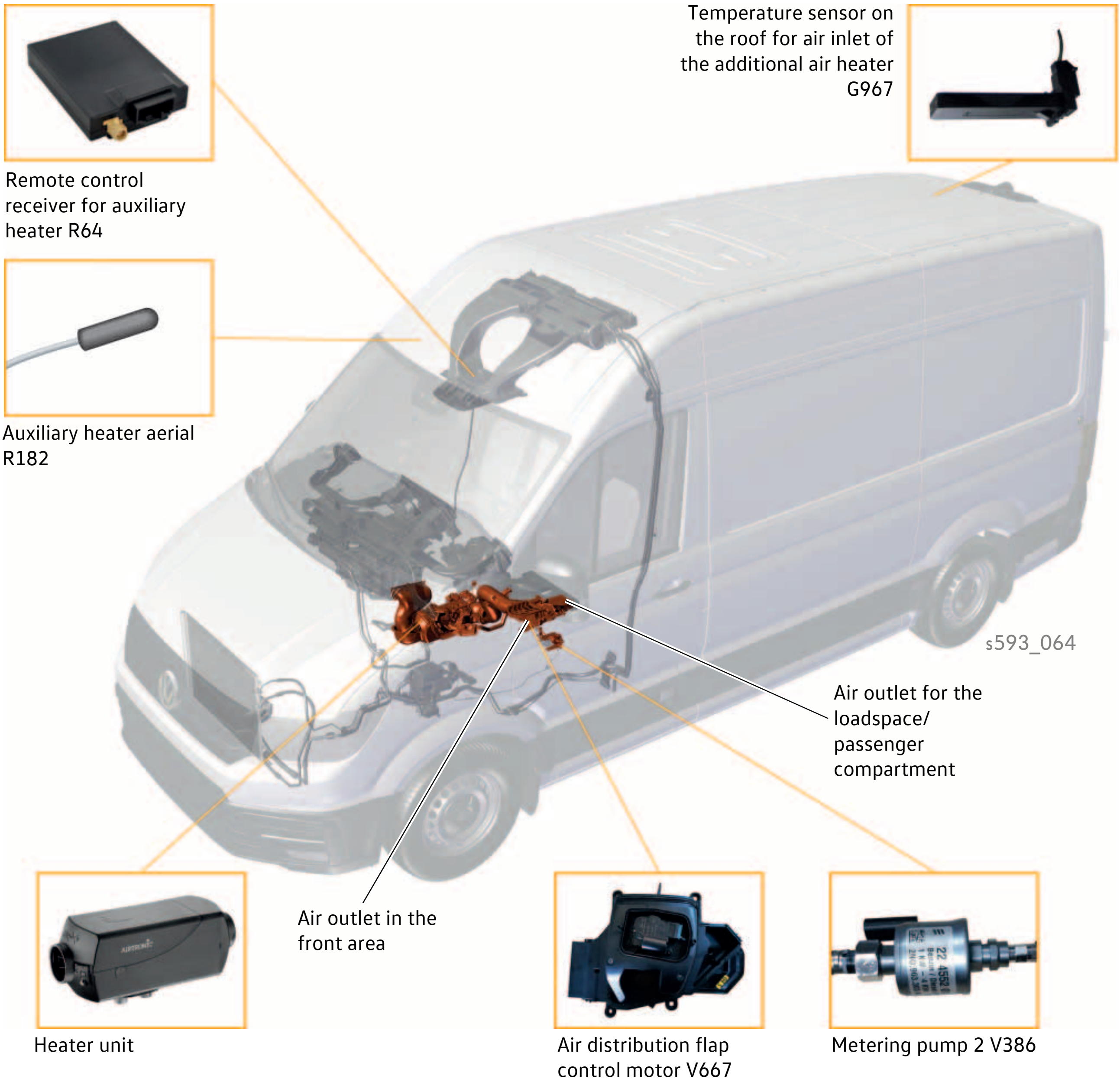
Technical data

s593_060

Manufacturer	Eberspächer
Designation	Airtronic M2 D4A
Use	For use on Crafter panel vans, platform vans with single cab, platform vans with double cab
Fuel	Diesel as per EN 590
Heating output	2 regulation stages • Power stage 3500 watts • Low 1000 watts
Fuel consumption	approx. 0.4 l/h max.
Operating voltage	12 volt
Electrical power consumption	45 watts max.
Undervoltage disconnection	10.5 V > 20 seconds

Supplementary equipment

Fitting locations of components



The **heater** of the additional air heater with control unit for additional air heater J604 is installed on the underbody on the right.
The **metering pump V386** is located on the underbody near the air distribution flap and supplies the additional air heater with the required fuel. It is a reciprocating piston pump and is controlled in a cyclical manner.
The **remote control receiver for auxiliary heating R64** is installed at the front of the roof near the interior mirror.
The **control motor of the air distribution flap V667** is located on the underbody in the middle of the vehicle.
The **temperature sensor on the roof for air intake of the additional air heater G967** is located in the rear third of the vehicle. It determines the temperature for the loadspace and passenger compartment.

Heater unit system connections

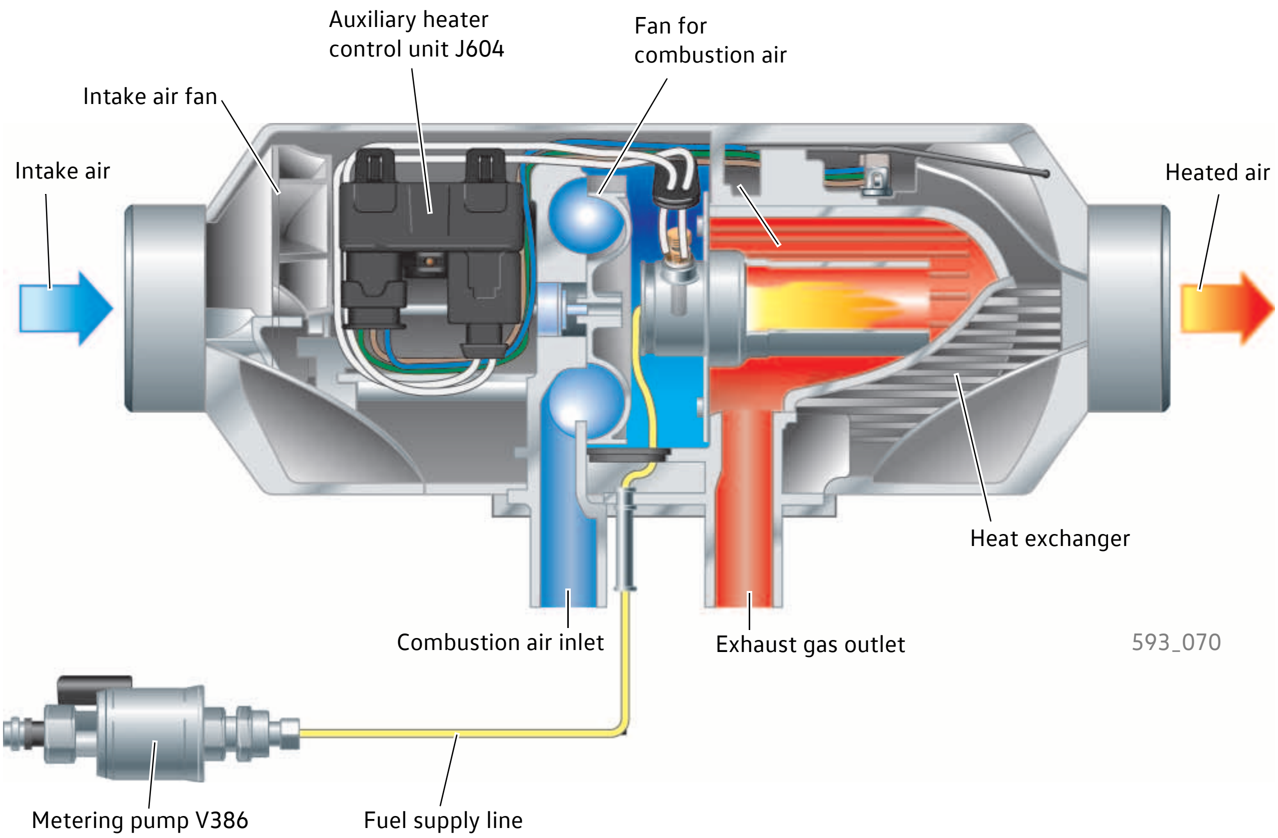
The central main component of the additional air heater is the heater. It has the following system connections:

Fuel system
Fuel is pumped by the metering pump V386 from the fuel tank to the heater via a separate line (fuel supply).

Intake and heated air
The intake air is drawn in via an aperture in the face end of the heater by the intake air fan of the combustion air blower V385 and is channelled past the heat exchanger at the outer side. The air absorbs heat whilst doing so, and emerges from an outlet aperture at the other face end.

Air supply for combustion
The air required for combustion is drawn into the heater by the intake air fan of the combustion air blower V385 via the combustion air inlet; from there, it is conducted into the combustion chamber.

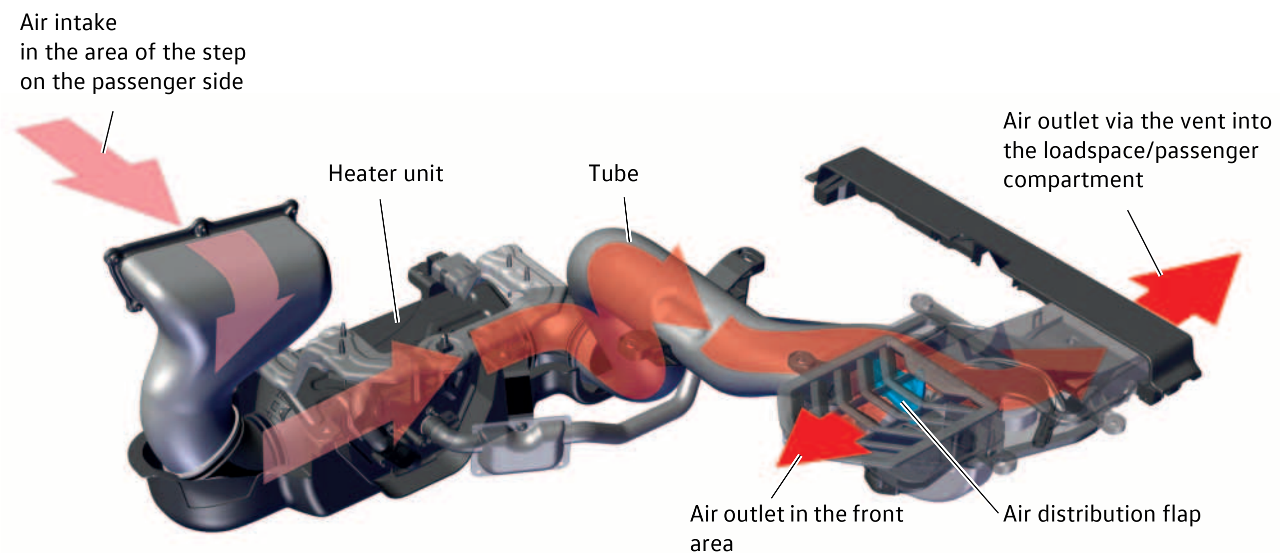
Vehicle electrical system
The additional air heater is connected to the electrical system via the electrical connections for the auxiliary heater control unit J604. The control unit is a component of the infotainment CAN bus.



593_070

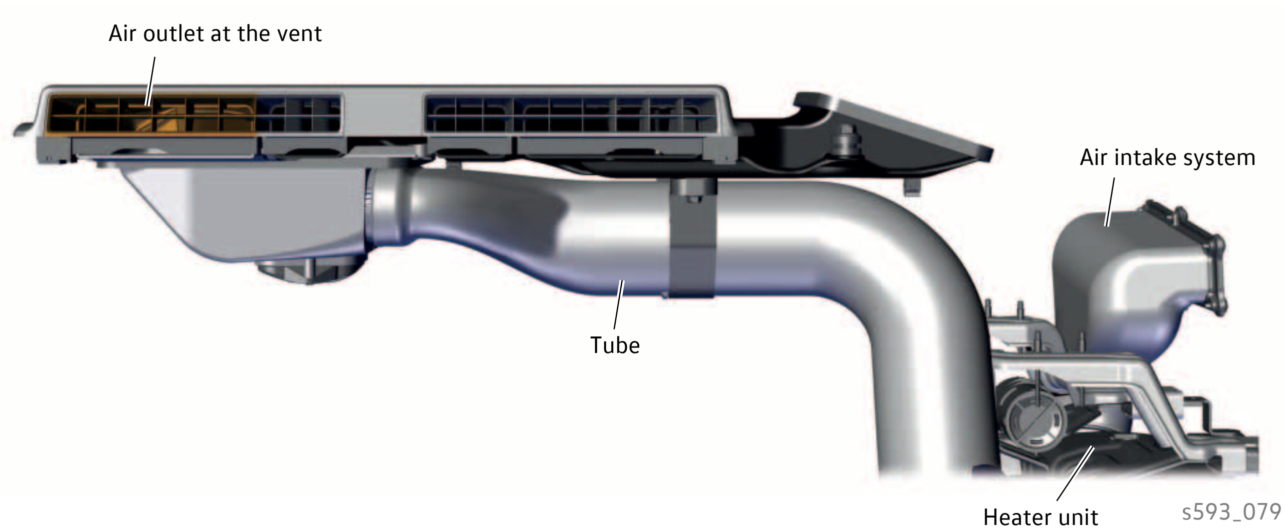
Air duct system

The air for the additional air heater is drawn in in the area of the step on the passenger side. The heater heats the air. A pipe channels the heated air to the air distribution flap. Depending on the position of the air distribution flap, the heated air is directed to the air outlet in the front area or to the vent in the loadspace/passenger compartment.



s593_063

Air is discharged from the vent into the loadspace/passenger compartment via the vent area marked in orange.

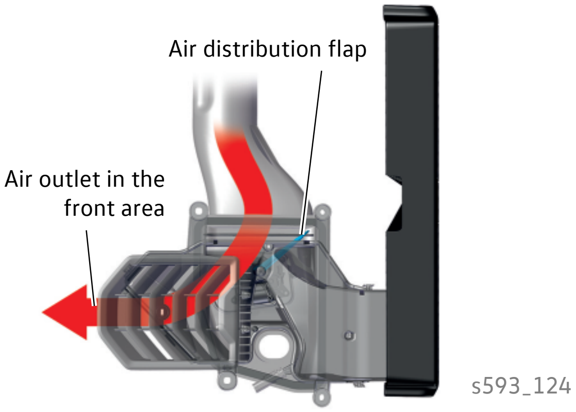


s593_079

Air distribution flap

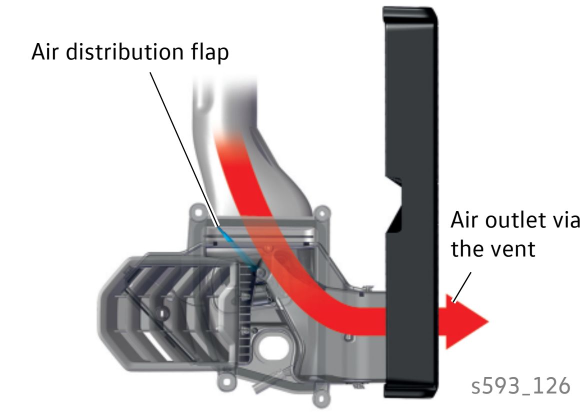
The control motor V667 can move the air distribution flap to three different positions.

Position 1



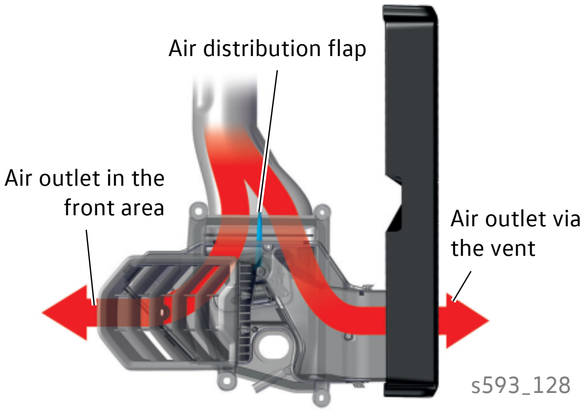
The air is only directed into the cab via the air outlet in the front area. The path to the vent in the loadspace/passenger compartment is closed by the air distribution flap.

Position 2



The air is only directed into the loadspace/passenger compartment via the vent. The path to the air outlet in the front area is closed by the air distribution flap.

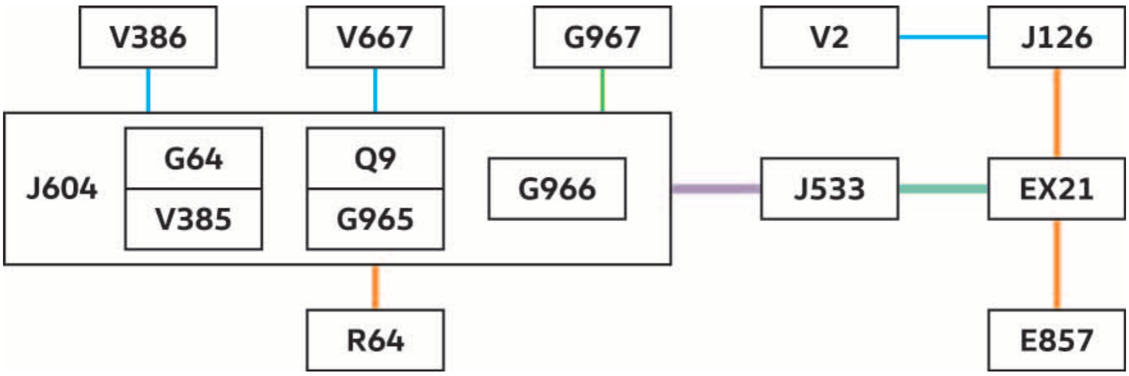
Position 3



The air is directed via the air outlet in the front area and via the vent into the cab or into the loadspace/ passenger compartment.

Supplementary equipment

Networking



s593_062

Key

E857 Additional display and operating unit 1
EX21 Heater and air conditioning controls
G967 Temperature sensor on the roof for air inlet of the additional air heater
J126 Fresh air blower control unit
J533 Data bus diagnostic interface
J604 Auxiliary air heater control unit
R64 Remote control receiver for auxiliary heater
V2 Fresh air blower
V667 Air distribution flap control motor 2
V386 Metering pump

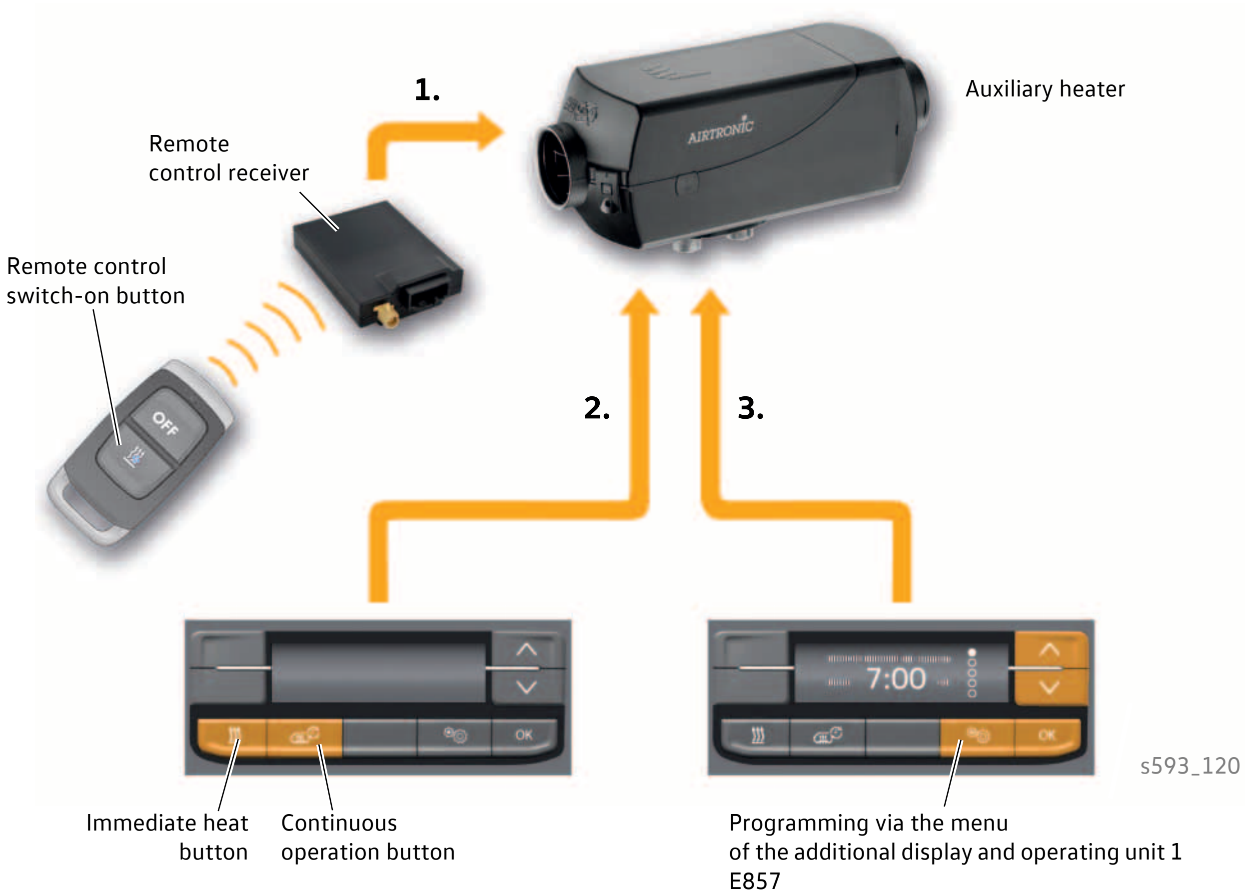
Internal components of the heater
G64 Flame monitor
G965 Temperature sensor for air inlet at the additional air heater
G966 Temperature sensor for air outlet at the additional air heater
Q9 Glow plug for heater
V385 Combustion air blower 2

Infotainment CAN bus (500 kbit/s)
Convenience CAN bus (500 kbit/s)
LIN bus wire
Sensor line
Actuator line

The climate control unit is the master for all functions in connection with the additional air heater.

Operation

The following variants are available for switching on the **auxiliary heater**:



1. Manually via the remote control
2. Manually via the immediate heat button or via the continuous operation button of the additional display and operating unit 1. (When the heater is switched on via the continuous operation button, the auxiliary heater runs for up to 48 hours.)
3. Automatically via pre-programming in the menu of the additional display and operating unit 1

If an additional water heater is additionally installed, the remote control receiver is connected there.

Supplementary equipment

Remote control

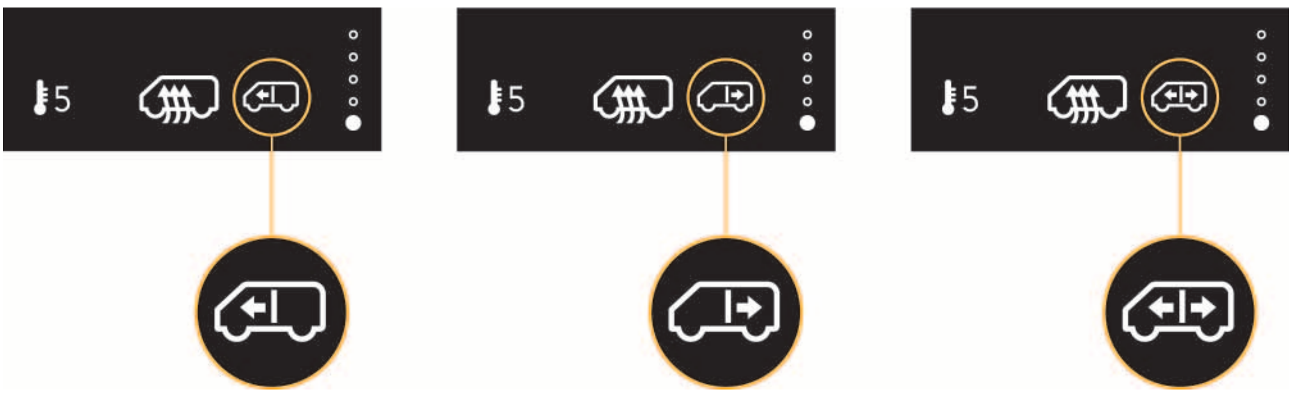
The auxiliary heater function of the additional air heater is operated via the remote control in exactly the same way as with the additional water heater. If an additional air heater and additional water heater are installed in the vehicle, then only the additional water heater is operated via the remote control.

Programming

The following programming for the auxiliary heating function of the additional air heater can be carried out via the buttons of the additional display and operating unit 1:

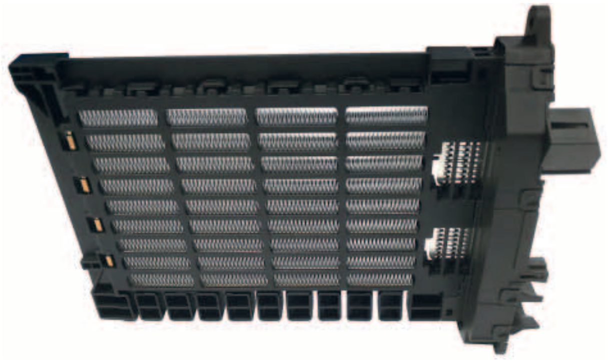
- Three different departure times (timers) can be programmed. The switch-on time and operating duration of the auxiliary heater is then determined by the onboard supply control unit according to the ambient temperature and the required interior temperature. There is no serial function of the timer, i.e. after the timer has expired, it must be selected again in order to use the set function again.
- You can program an operating duration between 10 minutes and 120 minutes in 10-minute steps.
- The heating levels can be set from 1 – H.
- The air flow direction can be adjusted. Under “Operating mode”, it is possible to select whether:
  - The air is only directed into the cab (Figure 1),
  - The air is only directed into the loadspace/passenger compartment (Figure 2),
  - The air is directed into the loadspace/passenger compartment and into the cab (Figure 3).

Figure 1 Figure 2 Figure 3



s593_130

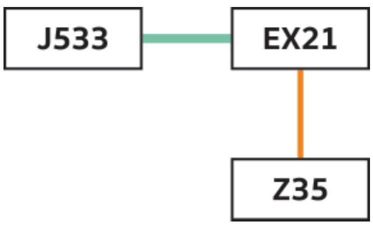
The electric additional air heater (PTC)



s593_121

The electric additional air heater (PTC) is installed directly downstream of the heat exchanger in the heater and air conditioning unit. It works as an air-based supplementary heater and immediately heats the cold air flowing into the vehicle interior during a cold start.

The additional air heater (PTC) is a purely electrically operated heating element with an output of 1400 watts. The PTC heating element is switched on by the heater and air conditioning controls EX21 via LIN-BUS.



s593_068

Key

EX21	Heater and air conditioning controls
J533	Data bus diagnostic interface
Z35	Auxiliary air heater element
(green line)	Convenience CAN bus (500 kbit/s)
(orange line)	LIN bus wire

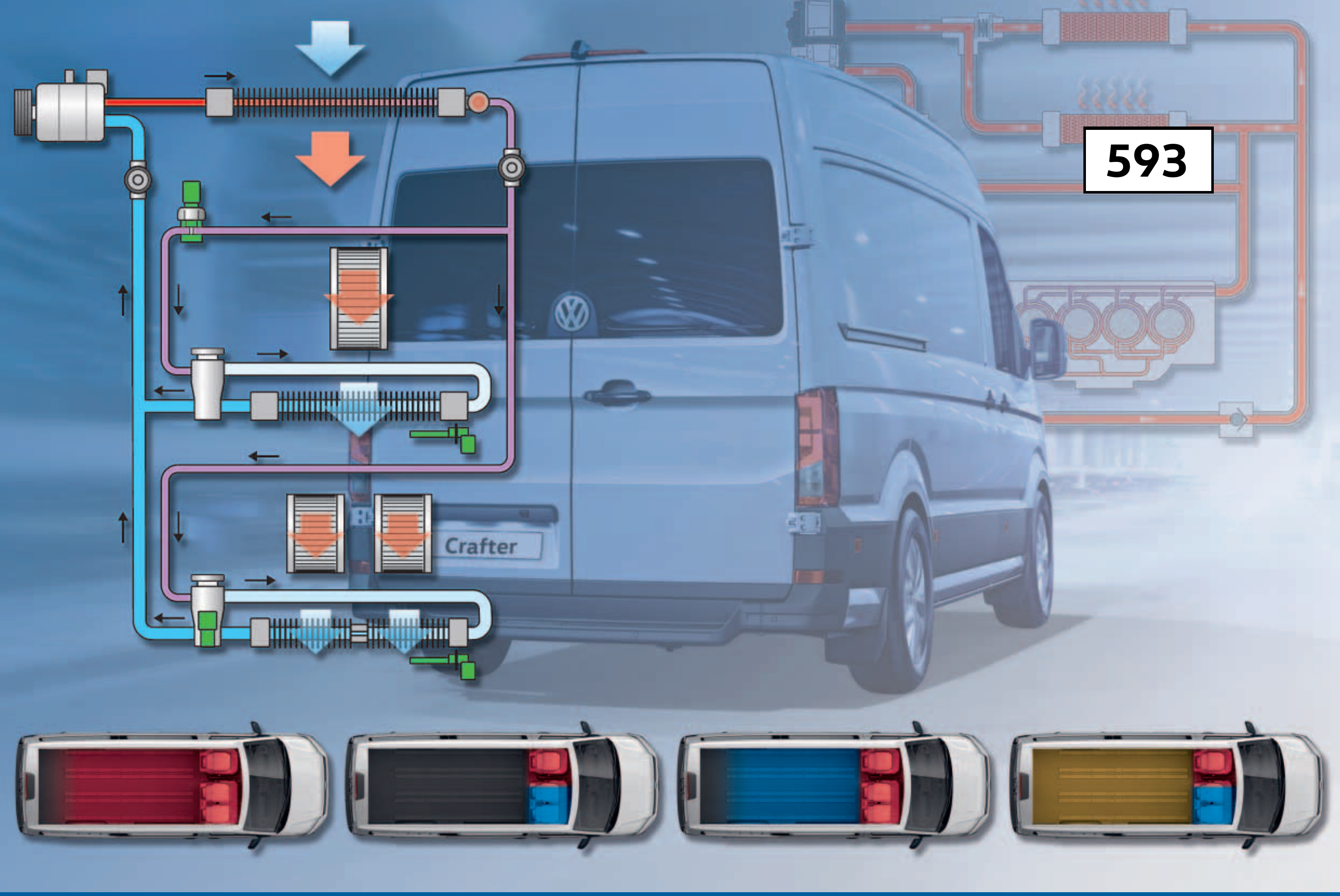
The additional air heater (PTC) operates without the driver's influence and only when the engine is running.

Switch-on conditions

- Switch-on request by climate control unit is present.
- Terminal 15 ON
- Ambient temperature approx. +4 °C
- Coolant temperature less than 80 °C
- Engine running time longer than 8 s
- Engine speed greater than 400 rpm
- Battery voltage > 12.2 V

Deactivation conditions

- At least one of the switch-on conditions is not fulfilled.
- Alternator defective or fully loaded
- Ambient temperature sensor fault
- Shutdown by load management



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